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Comparative Analysis of Uruk Period Ceramic Assemblages in Northern Syria

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Abstract

The Uruk Expansion is one of the defining phenomena of the Uruk period, marked by the spread of Uruk-style material culture across Greater Syro-Mesopotamia. The Uruk World System has long provided the most influential framework for explaining this phenomenon, though it has faced increasing criticism and the rise of alternative approaches offering new insights. Yet, regardless of perspective, major sites within this framework are still often discussed together without sufficient differentiation. A fundamental question therefore arises: are these sites truly identical in nature?

To address this question, this dissertation examines two Uruk enclaves in northern Syria, Jebel Aruda and Habuba Kabira-Süd, through the lens of ceramic analysis. The availability of the most recent excavation reports, published in 2014, 2021, and 2023, makes such an investigation possible. The results challenge prevailing narratives by showing that neither site was a "pure" Uruk settlement nor fully homogeneous. Typological, frequency, and concentration analyses demonstrate that while the two sites share the majority of ceramic forms, they differ in frequency and distribution. Moreover, both sites produced unique ceramic forms and maintained connections with local Chalcolithic traditions.

These findings call for a more nuanced understanding of Uruk sites, viewing them not as uniform extensions of southern Mesopotamia but as distinct communities that interacted with local Chalcolithic cultures in diverse ways.

Acknowledgements

Firstly, I would like to express my deepest gratitude to my supervisor, Philip Graham. It was through his teaching that I first encountered the topic of the Uruk Expansion, and under his guidance I developed a strong interest in these ceramics. Our weekly meetings have always been a source of fruitful discussion and valuable knowledge, and his support has been of immense importance to me.

Secondly, I extend my sincere gratitude to all the lecturers whose generosity, guidance, and care have been a cornerstone of my journey. Their profound wisdom and insights have granted me a deeper appreciation for humanity's brilliant civilizations and the discipline of archaeology. I also wish to thank my family and friends, whose encouragement has given me the strength to persevere.

Finally, I extend my most sincere appreciation to the excavators of Jebel Aruda and Habuba Kabira-Süd, as well as to all those who carried out fieldwork at these sites. Without their dedicated efforts and meticulous records, no excavation reports would have been published, and my own research would not have been possible. Their work ensures that the achievements of human history are preserved rather than lost to time. Looking ahead, I aspire to contribute to this field of archaeology myself, and to advance our understanding of the Uruk Expansion through future research.

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List of Abbreviations

BRB: Bevel-Rim Bowl

BT: Blumentopfe (Flowerpot)

Cat.: Category

DBI: Distribution Breadth Index

DII: Dispersion Intensity Index

FG: Formengruppen (Form Group)

HK: Habuba Kabira-Süd

JA: Jebel Aruda

1. Introduction

1.1 Background

The second half of fourth millennium BC in Mesopotamia represents one of the most transformative periods in human history. During this era, the long-standing Ubaid tradition (5500-4200 BC), which had shaped the greater Mesopotamian region over a millennium, gave way to a new cultural horizon of the Uruk period (4150-3100 BC) (Akkermans 2003; see Table 1). Named after the city of Uruk, the most prominent urban center in the southern alluvium of the time, the Uruk period witnessed great achievements of human civilization: the emergence of the first cities, the construction of monumental public architecture, and the invention of writing, administration, and bureaucratic systems.

Table 1: Chronological Framework of Syro-Mesopotamia (After Rothman 2001: 9)

Chronology	Northern Mesopotamia	Southern Mesopotamia	Santa Fe Conference Periodization
3000 BC	Early Bronze I	Jemdet Nasr	
		Late Uruk	LC5
			LC4
		Middle Uruk	
			LC3
4000 BC		Early Uruk	
			LC2
	Late Chalcolithic		
		Terminal Ubaid	LC1
5000 BC	Ubaid	Ubaid	

The Uruk period is conventionally divided into three sub-phases: Early, Middle, and Late Uruk. The Middle and Late Uruk period (ca. 3800-3100 BC) are distinguished by the phenomenon known as the "Uruk expansion". Over the course of this 600-year phenomenon, Uruk-related material culture including pottery, glyptic art, accounting tools, and architectural styles spread across greater Mesopotamia with varying intensity (see Figure 1). The expansion covered an estimated 5,500 km², incorporating hundreds of settlements throughout the Syro-Mesopotamian region (Stein 1999). Major sites include, but are not limited to: Tell Brak and Tell Hamoukar along the Khabur River; Nineveh and Tell Hawa along the Tigris River; the Jebel Aruda/Habuba Kabira-Süd/Sheikh Hassan cluster in the Tabqa Dam area; Carchemish and Hacınebi along the Euphrates River; Arslantepe and Samsat in southeastern Anatolia; and Godin Tepe in the Zagros region (see Figure 2). For unknown reason, the Uruk expansion collapsed abruptly around the end of the 4th millennium BC. Sites such as Jebel Aruda were completely abandoned, while others, like Tell Brak, underwent a clear cessation of material cultural connections with southern Mesopotamia (Schwartz 2001).

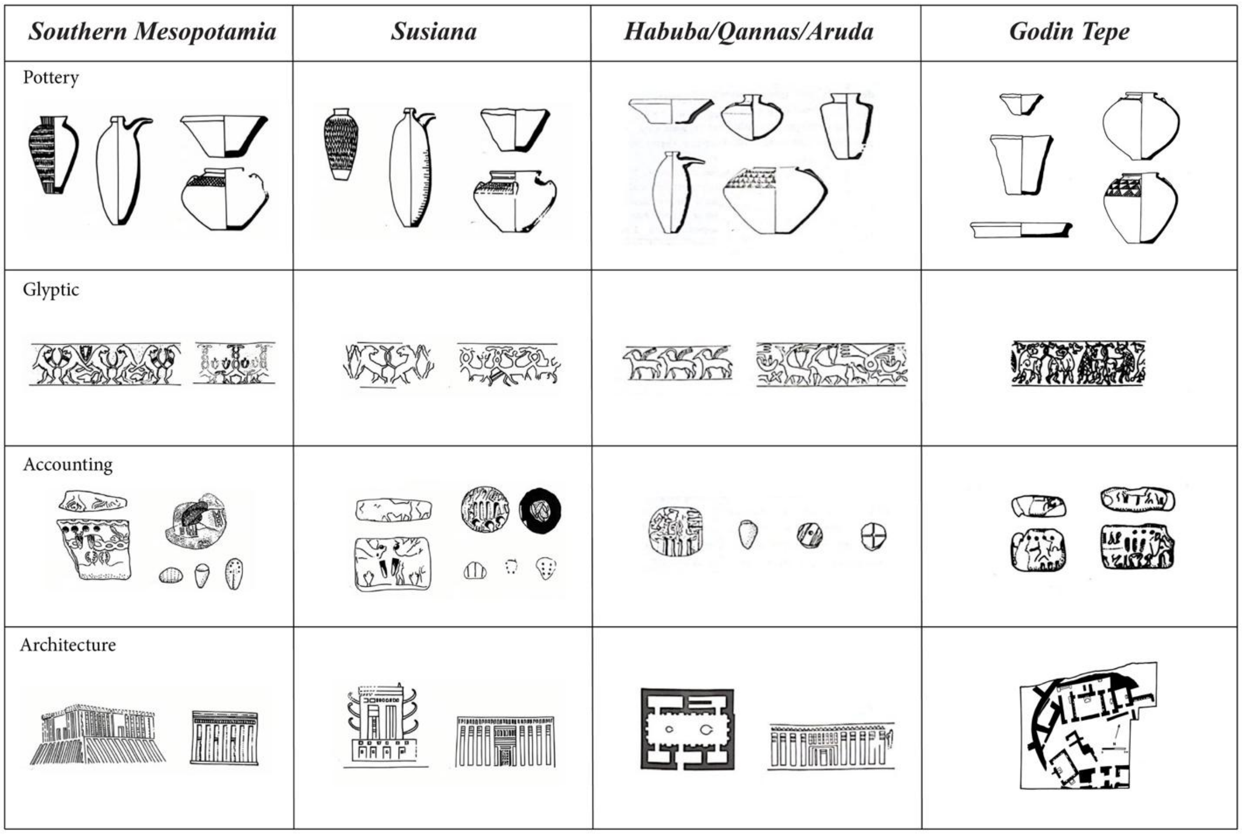
Southern Mesopotamia
Susiana
Habuba/Qannas/Aruda
Godin Tepe
Pottery
Glyptic
Accounting
Architecture

Figure 1: Selective Archaeological Findings from Major Sites of Uruk Expansion (Edited Based on Algaze 2004: Figure 17)

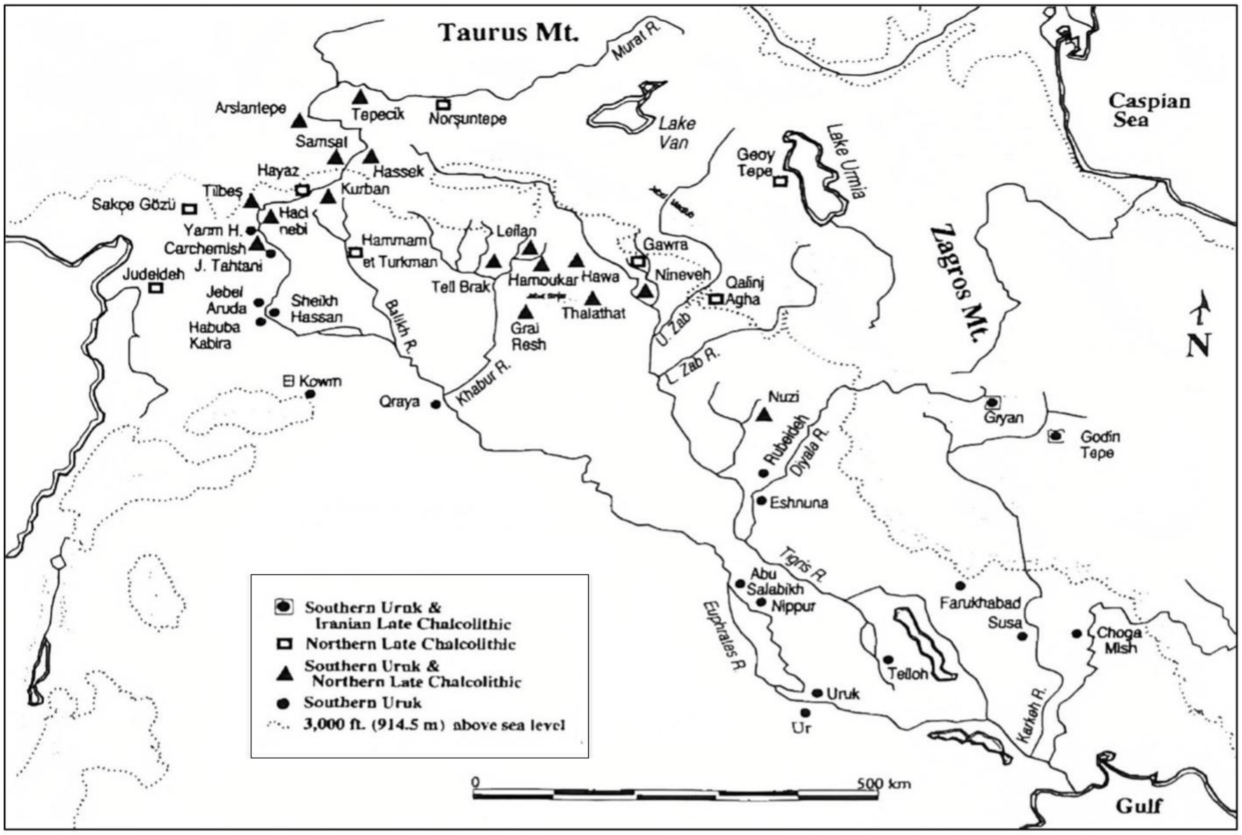
Taurus Mt.
Murat R.
Arslantepe
Tepecik
Norşuntepe
Samsat
Hassek
Hayaz
Kurban
Tilbeş
Sakçe Gözü
Haci nebi
Yarım H.
Carchemish
J. Tahtani
Judeideh
Jebel Aruda
Habuba Kabira
Sheikh Hassan
Hammam et Turkman
Balikh R.
Leilan
Tell Brak
Hamoukar
Hawa
Grai Resh
Thalathat
Khabur R.
El Kowm
Qraya
Lake Van
Lake Urmia
Geoy Tepe
Gawra
Nineveh
Qalinj Agha
U. Zab
L. Zab R.
Zagros Mt.
Caspian Sea
N
Nuzi
Rubeideh
Diyala R.
Gryan
Godin Tepe
Eshnuna
Tigris R.
Abu Salabikh
Nippur
Euphrates R.
Farukhabad
Susa
Choga Mish
Telloh
Uruk
Ur
Karkeh R.
Gulf
0
500 km
Southern Uruk & Iranian Late Chalcolithic
Northern Late Chalcolithic
Southern Uruk & Northern Late Chalcolithic
Southern Uruk
3,000 ft. (914.5 m) above sea level

Figure 2: Major Sites During Uruk Expansion (After Rothman 2001:6)

The complexity of this phenomenon has spurred significant scholarly debate regarding the nature of the Uruk expansion. Hypotheses range from the settlement of southern Mesopotamian refugees fleeing political instability (Johnson 1988) to the outcome of prolonged cultural interaction since the Ubaid period (Frangipane 2001). The most influential framework, however, is the Uruk World System proposed by Guillermo Algaze (Algaze 2004) , who interprets the expansion as an informal imperial network dominated by southern Mesopotamian city-states for resource procurement. According to this model, southern Mesopotamians, facing resource scarcity in their homeland, established enclaves, stations, and outposts at strategic communication nodes in the northern periphery to facilitate trade and economic activities. Typical Uruk enclaves include sizeable settlements with a full range of Uruk materials, such as the Carchemish complex, the Jebel Aruda /Habuba Kabira-Süd/Tell Qannas complex, Tell Brak, and Nineveh. Stations and outposts are generally smaller Uruk-related sites such as Tell Qraya, Tell Ramadi, Tell Rawa, Hassek Höyük, and Godin Tepe. Together, these enclaves, stations, and outposts formed a multi-tiered communication network, each playing distinct roles in facilitating long-distance trade along the Euphrates and Tigris rivers.

While this framework provides a compelling explanation for the Uruk expansion, it has faced increasing scholarly criticism. Recent archaeological excavations in northern Iraq, northern Syria, and southeastern Turkey (Ur 2010; Stein et al. 1996; Frangipane 2002) have revealed large indigenous buildings and structures, demonstrating significant social complexity in northern societies prior to Uruk contact. These findings challenge the passive and peripheral role assigned to northern communities in the Uruk World System and instead support a more balanced north-south relationship, as proposed in Stein's trade-diaspora theory. Based on evidence from Hacinabi Tepe, a newly identified Uruk-related outpost where intrusive Uruk materials coexist with indigenous cultural elements, this alternative model interprets Uruk expansion as the result of dispersed Uruk merchant groups engaging with autonomous host communities (Stein 1999).

Despite these advancements, the Uruk expansion remains a complex and enigmatic process without a definitive explanatory framework. While Stein's trade-diaspora model provides a more nuanced perspective on peripheral settlements, the Uruk World System continues to be the predominant paradigm, as no alternative model fully accounts for the multifaceted nature of the Uruk expansion.

1.2 Research Gaps and Objectives of the Study

While the Uruk expansion and the associated Uruk world system have ignited extensive scholarly research, two significant methodological challenges persist within the ongoing debate. First, interpretations often rely heavily on preliminary archaeological reports, which may lack the depth required for robust analysis. Second, the prevailing theoretical discourse, whether advocating for the Uruk World System or trade-diaspora model, frequently treats sites within categorical labels such as "enclave," "station," and "outpost" as internally consistent and undifferentiated. These sites are often discussed collectively, with minimal differentiation of their unique characteristics (Lupton 1996; Algaze 2004; Rothman 2001; Akkermans 2003) As a result, a fundamental question remains underexplored: to what extent are the sites implicated in the Uruk expansion homogeneous in their nature and function?

Recent comprehensive publications on Jebel Aruda and Habuba Kabira-Süd (Strommenger et al. 2014; Van Driel+ and Van Driel-Murray 2023) in northern Syria now provide detailed datasets, enabling a more nuanced examination of these colonies and facilitating investigation into this key question. Their geographical proximity within the Tabqa Dam area also minimizes environmental variables, creating an ideal controlled comparison. Analyzing these two sites together can therefore yield refined insights into the diversity of colonial practices during the Uruk period.

Although a comprehensive examination of all site artifacts is impractical, this study focuses on pottery assemblages as a proxy for comparing similarities and differences between the sites. Through this lens, the paper investigates the following questions and aims to contribute to a deeper understanding of the Uruk expansion:

1. What characterizes the pottery assemblage of an "Uruk enclave" in northern Syria during the Uruk expansion? How does this pottery inform our understanding of daily life at these sites?
2. How do contemporary Uruk-related sites compare? Are they identical, or do they exhibit notable differences? Can pottery analysis reveal insights into the specific nature of each site and the broader dynamics of the Uruk expansion?

1.3 Methodology and Dissertation Structure

This dissertation systematically investigates the pottery assemblages from the archaeological sites of Jebel Aruda and Habuba Kabira-Süd, focusing on their form, frequency, and spatial distribution. The study draws on data from the most recent published excavation reports: *Ausgrabungen in Habuba Kabira-Süd* (Strommenger et al. 2014; Kohlmeyer and Ludwig 2021) and *Jebel Aruda* (Van Driel+ and Van Driel-Murray 2023). These sources provide detailed ceramic descriptions, illustrations, and inventory lists that form the basis for the comparative analysis. The dissertation begins with Chapter 2, which provides an overview of the excavation history and pottery research at both sites, introducing their archaeological contexts and the typology study that underpin the subsequent analyses. The formal comparative study of pottery is conducted in Chapters 3, 4, and 5. Chapter 3 examines pottery morphology through a manual comparison of forms, based on descriptions and illustrations provided in the excavation reports. Chapter 4 undertakes a quantitative assessment of ceramic type frequencies, utilizing statistical data extracted from the inventory lists (Strommenger et al. 2014: 87–112; Van Driel+ and Van Driel-Murray 2023: 13–31) of both sites. Chapter 5 explores spatial distributions and patterns within the assemblages, employing Python-based analytical tools to process and interpret provenanced inventory data. Each of these analytical chapters includes a detailed methodology to ensure transparency and replicability. Finally, Chapter 6 synthesizes the findings from the comparative analyses, presenting a comprehensive summary of the results and addressing the research questions posed at the outset of the study.

2. Research Context: Sites and Typology

Both Jebel Aruda and Habuba Kabira-Süd are located in the Tabqa Dam area of northern Syria. The sites lie close to one another on the western bank of the Upper Euphrates, with Jebel Aruda situated about 8 km to the north (see Figure 3). Excavations at both settlements have been extensive, making them among the most thoroughly investigated Uruk enclaves and providing a detailed picture of their archaeological record. The following sections offer a brief overview of their excavation history and pottery research.

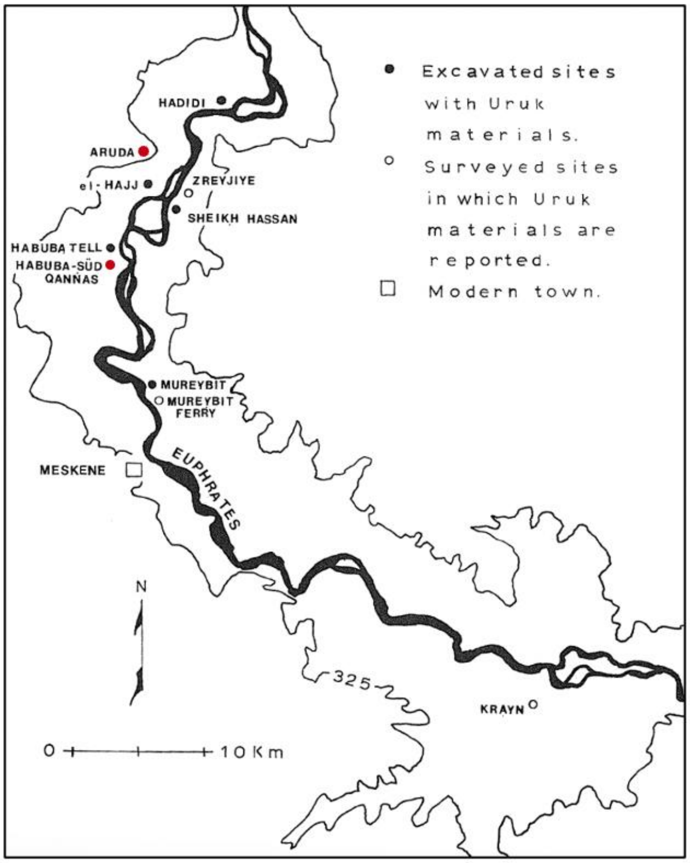
● Excavated sites with Uruk materials.
○ Surveyed sites in which Uruk materials are reported.
□ Modern town.
HADIDI
ARUDA
el-HAJJ
ZREYJIYE
SHEIKH HASSAN
HABUBA TELL
HABUBA-SÜD
QANNAS
MUREYBIT
MUREYBIT FERRY
MESKENE
EUPHRATES
N
325
KRAYN
0
10 Km

Figure 3: Jebel Aruda, Habuba Kabira-Süd and Other Uruk Related Sites in the Tabqa Area of Northern Syria (Edited Based on Algaze 2004: Figure 7)

2.1 Excavation History and Site Description

2.1.1 Jebel Aruda

Located on a 60m-high spur overlooking Euphrates River (see Figure 4), Jebel Aruda was first identified during M.N. van Loon's 1964 survey. Preliminary soundings were conducted in 1972-1974 as part of the project on the medieval settlement of Ta'as. The site's significance became clear in 1975 with the discovery of massive walls in the temple area, prompting systematic excavations led by Govert van Driel's Dutch team in 1977, 1978, and 1982. Funding constraints halted work in 1982, and the final report was published in 2023, commemorating van Driel's passing in 2002 (Van Driel+ and Van Driel-Murray 2023).

Excavations revealed that the site was founded on virgin soil and was never reoccupied, except for minor Byzantine activities. Constrained by the wadi to the west and the upper edge of the Jebel to the south, the preserved area measures 290 m at its greatest length and 230 m at its greatest width. Two test pits dug across the wadi during the 1982 campaign uncovered decayed mudbrick and collapsed wall fragments, suggesting the settlement may have extended further west and covered at least 4 hectares (Vallet 1998). Following the elevation gradient, the site was divided into three areas: the northern house area with 7 units (Houses NA to NH) situated along the 346-350m contour lines; the central temple area positioned on the 350-357m contour lines; and the southern house area with 12 units (Houses SI to SVIII), of which SV, SVI, and SVII stood on a higher sub-terrace in the southernmost sector (see Figure 4).

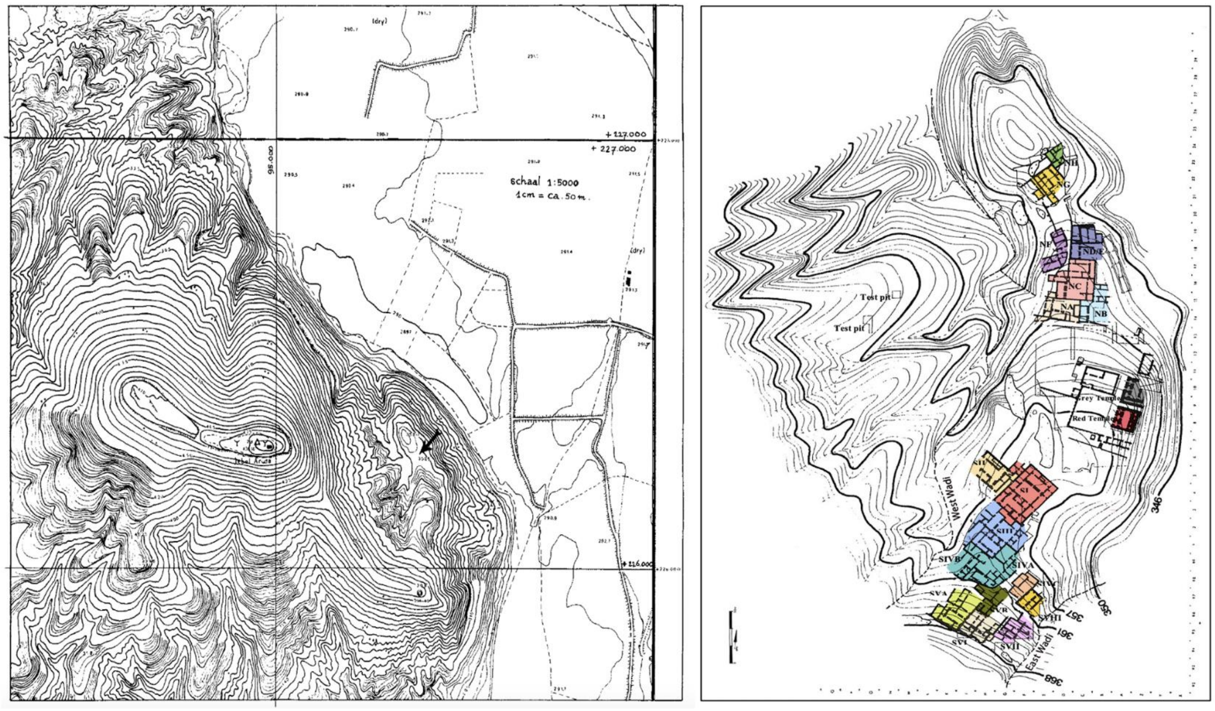
(dry)
+ 227.000
+ 227.000
95.000
Schaal 1:5000
1cm = ca. 50m.
(dry)
Jebel Aruda
+ 226.000
NH
NG
NF
ND/E
NC
NA
NB
Test pit
Test pit
Grey Temple
Red Temple
West Wadi
SII
SI
SIII
SIVB
SIVA
SIVC
SVA
SVB
SVIII
SVI
SVII
East Wadi
346
350
357
361
368

Figure 4: Map (left) and Plan (right) of Jebel Aruda (Edited Based on Vallet 1998: Figure 1; Van Driel+ and Van Driel-Murray 2023: Figure 1.4 and Figure 1.7)

The site underwent three distinct phases, most clearly visible in the central temple area (see Figure 5). In Phase I, the Red Temple was constructed using Riemchen bricks made of red erosion material. A later subphase (Ib) saw the addition of a "lower kitchen" with hearths. During Phase II, the Grey Temple was built adjacent to the Red Temple using Riemchen bricks made from valley-sourced grey mud. The original lower kitchen was filled in and replaced by a similar structure to the north. The final Phase III featured the construction of a massive terrace, though erosion has obscured any structures that may have stood atop it. In the northern and southern quarters, the area experienced a similar process of modification and expansion, where the settlement appears to have expanded from core buildings such as SI/SII/SIII and NA/NB/NC to include NH in the north and complexes on the higher terrace, accommodating a growing population, before the site was destroyed by severe conflagration and ultimately abandoned.

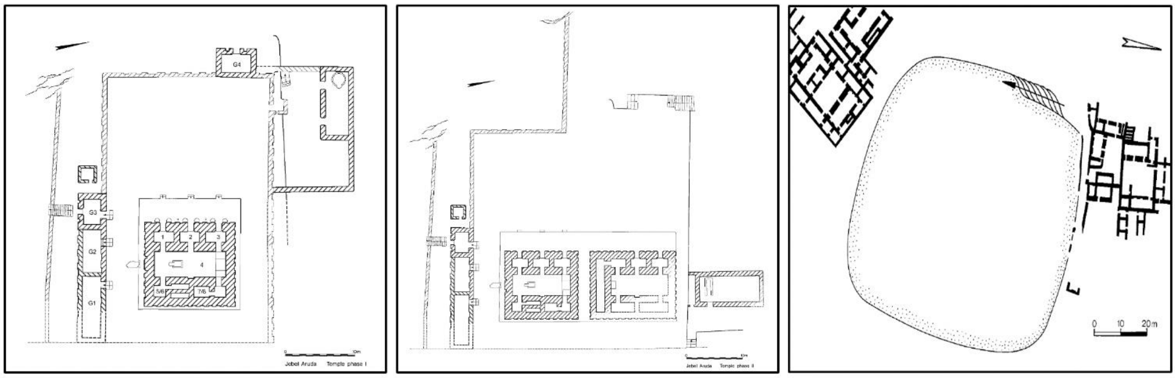

Figure 5: Architectural Evolution of Temple Area, Phases I, II, and III (After Van Driel+ and Van Driel-Murray 2023: Figure 2.7, Figure 2.17 and Figure 2.30)

Radiocarbon dating indicates the site dates to cal. 3333-3099 BC and cal. 3308-2917 BC, aligning well with the Late Uruk period. The excavation revealed a complete assemblage of Uruk-style pottery, tablets, and accounting tools stacked or scattered in rooms and on pathways, making this one of the best-preserved settlements in northern Syria. Based on the architectural remains and artifacts, the unwalled site has been interpreted as a temple complex with associated domestic structures (Van Driel+ and Van Driel-Murray 2023) and potentially a regional religious or administrative center (Strommenger 1980).

2.1.2 Habuba Kabira-Süd

Approximately 8 kilometers south of Jebel Aruda, the Uruk settlement of Habuba Kabira-Süd and its associated acropolis, Tell Qannas, lie along the Euphrates River (see Figure 6). Excavated by German and Belgian teams in response to the construction of the Tabqa Dam in the 1960s, the site was investigated in ten rescue campaigns by the German Oriental Society between 1969 and 1975, first under Ernst Heinrich and later Eva Strommenger.

Like Jebel Aruda, Habuba Kabira-Süd was established on virgin soil but on a substantially larger scale, extending over approximately 18 hectares. The settlement was occupied for no more than 150 years during the Late Uruk period before its abrupt abandonment. In later periods, the site functioned as a cemetery during the Early Bronze Age and Roman era. Excavations yield numerous Uruk-style ceramic vessels, including characteristic bevel-rim bowls (BRBs), flowerpots (Blumentöpfe, BTs) and spouted jugs. Typological analysis links these finds to Layer

17B in Susa Acropole I, Layer XV in Nippur, TW 12 in Brak, VIb-a in Uruk, and Layer 4 in Sheikh Hassan. (Strommenger et al. 2014).

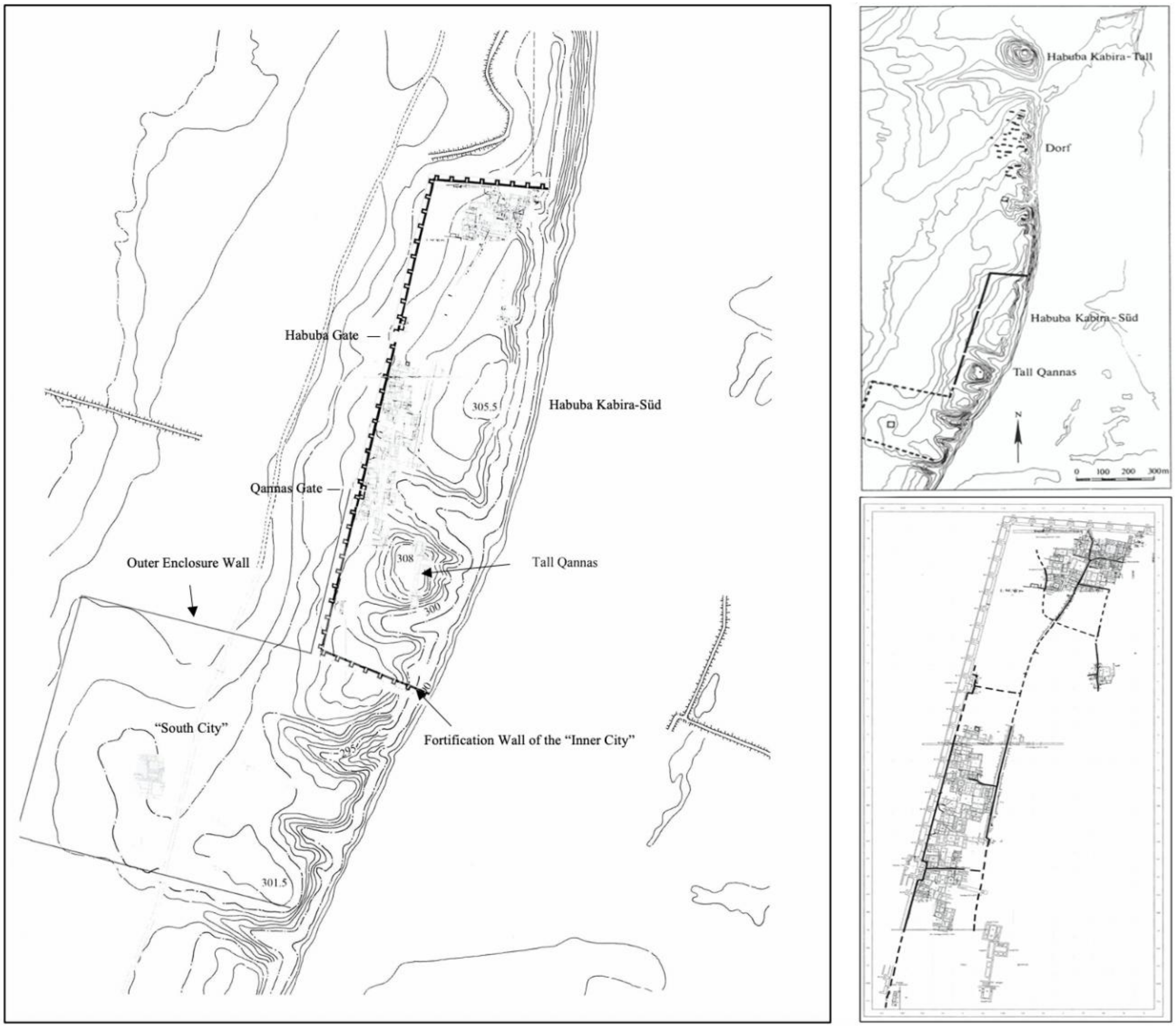
Habuba Gate
Habuba Kabira-Süd
305.5
Qannas Gate
Outer Enclosure Wall
308
Tall Qannas
300
"South City"
Fortification Wall of the "Inner City"
295
301.5
Habuba Kabira-Tall
Dorf
Habuba Kabira-Süd
Tall Qannas
N
0 100 200 300m

Figure 6: Map (upper right), City Plan (left) and Road System (lower right) of Habuba Kabira-Süd (Edited Based on Strommenger 1980: Figure 12, Kohlmeyer and Ludwig 2021: Plate 103 and Plate 94)

The city was constructed on a narrow Euphrates terrace, whose topography determines the city's main axis and house orientation (see Figure 6, upper right). It comprises an inner city, enclosed by a robust 3-meter-thick fortification wall, with residential quarters, workshops, and storage areas in the north, and an administrative or cultic center at Tell Qannas in the south. A southern settlement area, the "south city," remains poorly understood owing to erosion (see Figure 6, left). Both the inner city and south city were surrounded by a thinner outer wall. Three construction levels are attested, though the principal phase remains debated. The northeastern section appears to represent the earliest occupation and the fortifications were added later (Vallet 1996; Ludwig W. 1977).

The urban design of Habuba Kabira-Süd demonstrates advanced town planning, incorporating fortifications, a road network, and a drainage system. The inner wall, fortified with evenly spaced bastions, had two main gates: the Habuba Gate in the north and the Qannas Gate in the south. Within the wall, 56 house complexes were identified, many with gravel-paved courtyards and drainage channels. The road system comprised a central north-south "long street", a parallel street along the city wall, and several east-west streets connectors (see Figure 6, lower right). The drainage system, integrated into both streets and houses, efficiently channeled water toward the western slope. Collectively, these features provide some of the earliest evidence of Uruk period urban planning, supporting a peak population estimated at 6,000-8,000 (Strommenger 1980).

Frequently interpreted as a "trading colony" and "gateway city," Habuba Kabira-Süd likely controlled a pivotal trade route along the Euphrates, facilitating long-distance exchange with southern Mesopotamian cities. It is hypothesized that the city engaged in seasonal trade, with timber from the Amanus and Lebanese forests and salt from Lake Gabble brought in by cattle in the autumn. These goods, along with manufactured products such as leather, were then transported southward along the Euphrates in the spring (Strommenger et al. 2014).

2.2 Pottery Typology

2.2.1 Jebel Aruda

The pottery assemblage from Jebel Aruda, as documented in the excavation report's inventory list[1], comprises 1,681 registered items, including complete vessels (designated JA), fragmentary vessels (designated JAF), and unconnectable sherds (designated SH). These are classified into 44 typological groups (Cat.1-44) based on form, fabric, construction method, and function. Category 43 contains Uruk-period pottery that does not fit into established groups, while

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[1] Van Driel+ and Van Driel-Murray 2023, Book 2, pp. 13–31, Concordance II: By Find Number

Category 44 encompasses non-Uruk period material. The inventory records 216 BRBs and 124 BTs, although these figures underestimate their true frequency, since the abundance of unconnectable sherds prevents precise quantification. The assemblage is dominated by characteristic Uruk forms, notably BRBs, BTs, elongated spouted vessels, and pots with incised or false-spout decoration.

Fabric analysis distinguishes six groups, four of local origin and two probably non-local. The most prevalent fabric is a buff to pinkish riverine clay ("common ware"), typically poorly cleaned and tempered with chalk and fine sand, used for most Uruk-style forms. A finer variant of this fabric, ranging in color from white or grey to buff, was employed for small and delicate vessels. Another significant fabric is the well-burnished Uruk Red ware (Cat. 14), made of red clay with minimal inclusions and associated with higher-value or prestige items. By contrast, a coarse, chaff-tempered, low-fired ware was employed for thick-walled handmade cooking and storage vessels (Cat. 19 and Cat. 30). Among the non-local fabrics, one is a soft, polished buff ware with a soapy texture, used for "Uruk Red copies" (Cat. 13). Diatom analysis suggests these vessels may have been imported from southern Mesopotamia. The second non-local ware consists of thin, organically tempered blistered round-bottom pots (Cat. 29). These sherds are extremely hard and overfired, with greenish surfaces and visible air bubbles.

Technological and functional analyses reveal a wide spectrum of manufacturing methods and vessel uses. Handmade techniques such as pinching, coiling, and slab construction were employed, alongside wheel-throwing, attested by rotational striations and string-cut bases. Notably, vessels within the same typological category often exhibit varied construction techniques, demonstrating flexibility in production practices. The pottery fulfilled diverse functions, ranging from practical daily activities such as eating and drinking, food preparation, cooking, storage, to specialized roles in rituals or as status symbols, reflecting the multifaceted needs of the community. Detailed functional attributions for each category are provided in the excavation report and are summarized in the table below.

Table 2: Pottery Categories by Function

Function	Category
Eating and drinking	3/4/5/7/9/(1/2)
Preparation of food	1/2/6/15/16/19/36/37/(38)/(17/19/20)
Bulk storage	28/33/34/35/27
Storage (for immediate use)	17/18/20/21/22/24/27/28/30
Storage of liquids	23/25/26/(38)
"Luxury" wares	10/11/13/14/39/40/(8/12)
Other	43

2.2.2 Habuba Kabira-Süd

At Habuba Kabira-Süd, the German excavation team employed a meticulous, sherd-based methodology for ceramic analysis, systematically documenting attributes such as fabric composition, surface treatment, and vessel morphology. This approach contrasted with the vessel-oriented methodology adopted by the Dutch team at Jebel Aruda. The recorded material includes 94,754 diagnostic sherds, alongside BRBs and BTs. Both connected and disconnected sherds are cataloged as entries G1-1007 in the inventory list[2]. The ceramics were classified into 88 typological groups and subgroups (FG 1-67, including 21 subgroups) distributed across 26 distinct fabrics (Ware 1-26). This classification prioritizes general vessel forms, with further differentiation based on rim and base characteristics. Additionally, 23 surface treatments were identified, comprising 14 techniques applied to the entire surface (such as smoothing and engobing) and 9 techniques restricted to partial surface modification (including grooving and incision). Consistent with findings at Jebel Aruda, BRBs and BTs dominate the assemblage, supplemented by other diagnostic Uruk-period forms such as spouted vessels and trays.

Based on fabric characteristics and manufacturing techniques, the pottery assemblage (FG 1-67) was further incorporated into five main categories: Groups I-IV and a collection of misfired pottery designated as Group FB. Sherd counts, excluding BRBs and BTs, indicate that Group I constitutes 83.72% of the diagnostic assemblage. This group consists of pottery made from standard Wares 1-3, produced with purified clay from river terraces and alluvial deposits. The clay contained lime and iron, with or without added river sand or quartz temper. Most vessels were wheel-thrown and fired at temperatures between 600 and 900°C. A smaller proportion of the pottery (8.47%) comprised Group II. These vessels were made from uncleaned clay rich in lime and iron and fired at 500-800°C. Group III accounted for only 0.46% of the assemblage and consisted of highly calcareous, iron-rich uncleaned clay fired at 400-800°C. In contrast to these Uruk-affiliated groups, Group IV (6.45%) represented a distinct tradition that shows clear connections with local Late Chalcolithic pottery of the late Amuq F and early Amuq G phases. This group consisted largely of handmade vessels produced from uncleaned, chaff-tempered clay and fired at considerably lower temperatures (350-500°C). These vessels stood in sharp contrast to typical Uruk-style pottery. Group IV also includes three imported examples, potentially originating from Anatolia, Palestinian regions, or other Late Chalcolithic contexts.

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[2] Strommenger et al. 2014, pp. 87-112, Inventarliste I

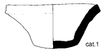
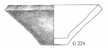
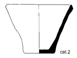
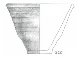
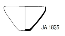
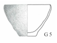
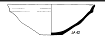
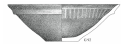
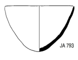
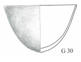
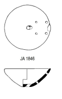
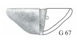
3. Morphological Comparison

With the research context established in the previous chapter, the systematic examination begins here. This chapter presents a morphological comparison of the ceramic assemblages from Jebel Aruda and Habuba Kabira-Süd. The study was conducted on 41 categories from Jebel Aruda and 66 FG groups and their sub groups from Habuba Kabira-Süd. Categories 44 (non-Uruk period material), 31 and 32 (both comprising unclassifiable sherds and rims) from Jebel Aruda and FG 54 (unidentifiable sherds) from Habuba Kabira-Süd are excluded from analysis. As the two assemblages were originally classified according to different systems (categories at Jebel Aruda and groups at Habuba Kabira-Süd), they were unified under a common "types" framework for comparison. The results of this comparative analysis are discussed below.

3.1 Common Forms Shared at Jebel Aruda and Habuba Kabira-Süd

The typological comparison reveals significant similarity between the assemblages. Jebel Aruda and Habuba Kabira-Süd share 58 common vessel forms, comprising 19 open (Types 1-19) and 39 closed shapes (Types 20-58) (see Table 3). This common repertoire aligns with typical Uruk-period pottery.

Table 3: Pottery Categories Attested in both Jebel Aruda (JA) and Habuba Kabira-Süd (HK)

Type	Category (JA)	Group (HK)	Description	Pottery in JA (not to scale)	Pottery in HK (not to scale)
1	Cat. 1	FG 19	Bevelled rim bowls (BRBs)		
2	Cat. 2	FG 14	Flowerpots (Blumentopfe, BTs)		
3	Cat. 3	FG 1	Simple conical goblets ("tea cups")		
4	Cat. 4	FG 8	Dishes with a flat base and a beaked rim[3]		
5	Cat. 5	FG 3.2 (Part of)	Small round-based bowls		
6	Cat. 6	FG 6	Small round or flat bottom bowls with holes on the walls and base ("spoon/ladles")		

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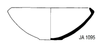
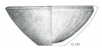
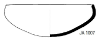
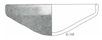
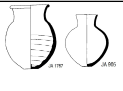
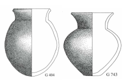
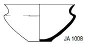
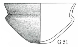
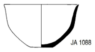
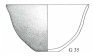
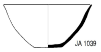
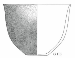
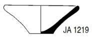
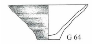
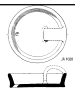
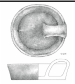
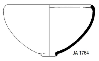
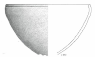
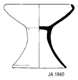
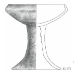
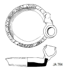
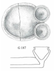
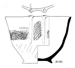
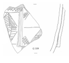
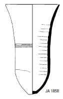
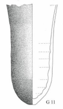
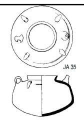
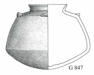
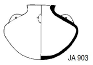
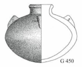
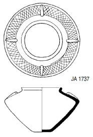
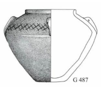
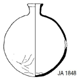
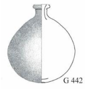
[3] JA contains three example of footing bowl, which is not attested in HK

7	Cat. 7 (Part of)	FG 11.2	Bowls with simple straight rims or beaked inturned rims [4]		
8	Cat. 7 (Part of)	FG 12.1	Lid?		
9	Cat. 8	FG 22.6	Sinuous or bottle-like shapes		
10	Cat. 9A	FG 3.4	S shaped bowls[5]		
11	Cat. 9B	FG 3.3	Small bowls with an angle on the wall[6]		
12	Cat. 9C	FG 10.2	Small bowls without an angle on the wall		
13	Cat. 9D	FG 5 (Part of) FG 7 (Part of)	Trumpit like small bowls		
14	Cat. 36	FG 15	Organic platter and basins		
15	Cat. 37	FG 11.4	Deep bowls with a central whole ("colander")		
16	Cat. 39	FG 12.4	Fruit stands, often red engobed		
17	Cat. 40	FG 13	Dishes with small cups on the rim		
18	Cat. 43	FG 10.1 (G109)	Decorated bowls with vertical lugs		
19	Cat. 43	FG 2 (Part of)	Carinated and flared beakers		
20	Cat. 10A	FG 44	Thin walled pots with shoulder lugs and rounded-flattish base, with or without false spout ("inkwell")		
21	Cat. 10B	FG 25	Pots with narrow neck, shoulder lugs and round base		
22	Cat.10C/10 D/ Cat. 27	FG 27	Pots with incised decoration and shoulder lugs		
23	Cat. 11	FG 24	Finely made bottles ("perfume bottle"?)[7]		

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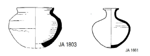
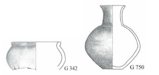
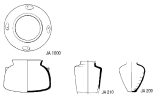
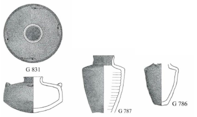
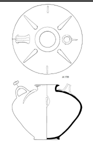
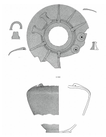
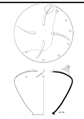
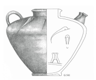
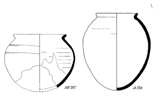
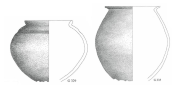
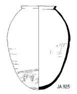
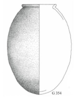
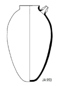
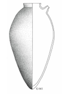
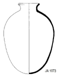
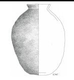
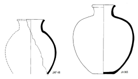
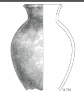
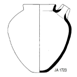
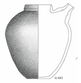
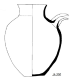
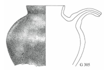
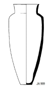
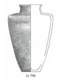
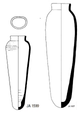
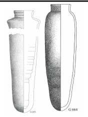
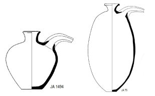
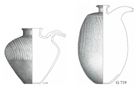
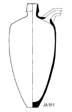
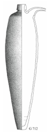
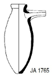
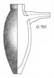
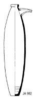
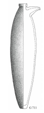
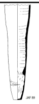
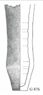
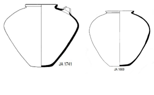
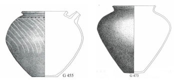
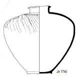
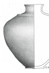
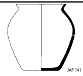
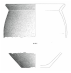
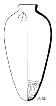
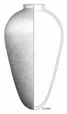
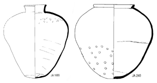
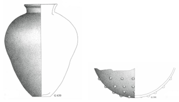
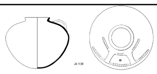
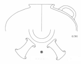
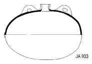
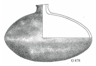
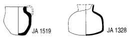
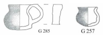
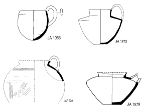
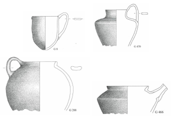
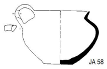
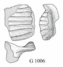
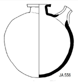
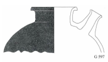
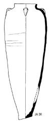
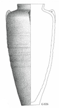
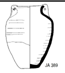
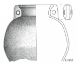
[4] HK contains one example of large footing bowl, which is not attested in JA

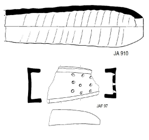
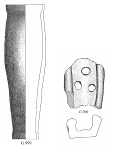
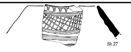
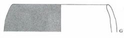
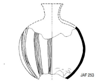
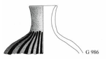
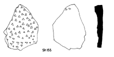
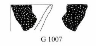
[5] HK contains one example with footing which is not attested in JA

[6] HK contains one example with footing which is not attested in JA

[7] Spouted forms are not attested in HK

24	Cat. 12	FG 22.4/ FG 22.6/ FG 40 (Part of)	Small closed shapes not belonging to Type 25 or Type 49		
25	Cat. 13	FG 44 / FG 41 (Part of)	Miscellaneous Uruk Red vessels		
26	Cat. 14	FG 31	Uruk Red vessels		
27	Cat. 28 (Part of)	FG 29	A group of 'Uruk Red copies' made of common ware		
28	Cat. 15	FG 22.3/ FG 22.4 (Part of)	Ordinary cooking pots		
29	Cat. 17	FG 22.4 (Part of)	Egg-shaped pots		
30	Cat. 18	FG 36 (G661)	Large egg-shaped vessels with spout and flattened base		
31	Cat. 20	FG 57	Baggy jars with a round base, flaring rim and smoothed exterior		
32	Cat. 21	FG 40?	Round based or bottle shape bottles		
33	Cat. 22	FG 37.1	Jars with short straight rims and straight spout on shoulder		
34	Cat. 23	FG 22.3 (G305)	Spouted vessels		
35	Cat. 24A	FG 41	Kink shouldered flat-bottomed vases		
36	Cat. 24B	FG 36	High, narrow, open bottles with thick round base		
37	Cat. 25	FG 39.2/ FG 32	Bottles with narrow mouth, drooping spout		
38	Cat. 26A	FG 39.1	Elongated bottle with folded rim and curving spout		

39	Cat. 26B	FG 39.1	Elongated bottle with collar like rim and straight spout		
40	Cat. 26C	FG 39.1	Elongated bottle with straight rim and drooping spout		
41	Cat. 26C	FG 45 (Part of)	Huge imcomplete elongated bottle		
42	Cat. 27	FG 26	Thin-walled storage jars with a flat base		
43	Cat. 28	FG 35.1	Storage jars with decorated shoulder		
44	Cat. 30	FG 58	Coarse storage jars		
45	Cat. 33	FG 36 (Part of)	Torpedo-shaped storage vessels		
46	Cat. 34	FG 35.2/ FG 22.5 (Part of)	Large storage jars, some with special surface treatment		
47	Cat. 35	FG 22.5 (G381)	Side rolled vessel		
48	Cat. 38	FG 46	Closed oval vessels with a narrow spout on the top		
49	Cat. 41	FG 22.1/ FG 22.2/ FG 32 (Part of)	Miniatures		
50	Cat. 42	FG 2/ FG 22.2/ FG 26 (Part of)	Vessels with handles and spouts		
51	Cat. 42	FG 66	Twisted handled vessels		
52	Cat. 43	FG 34	Black engobed pots		
53	Cat. 43	FG 43	Vases with cord loops		
54	Cat. 43	FG 45 (Part of)	Bottles with eyelet lugs on the rim and shoulder		

55	Cat. 43	FG 48 FG 49	Gutters and drains		
56	Cat. 43	FG 56	Hole mouth jars		
57	Cat. 43	FG 62	Painted small bottle		
58	Cat. 43	FG 67	Black sherds with white inlay		

3.1.1 Open vessels

Open vessels at the sites primarily consist of various bowls, cups, dishes, and trays. Among these, the most distinctive versions are BRBs (Type 1) and BTs (Type 2), which are prevalent at both Habuba Kabira-Süd and Jebel Aruda. The BRB, a hallmark of Uruk expansion, is particularly notable for its characteristic beveled rim, standardized form, and widespread distribution across Uruk-related sites (Nissen 1970). Its uniform size and evidence of mass production have prompted extensive scholarly debate regarding its function, with proposed uses including votive vessels, ration containers, and bread molds (Kaercher 2009). Alongside the mass-produced BRBs and BTs, other common forms include simple conical goblets (Type 3), beaked-rim dishes (Type 4), small round-based bowls (Type 5), bowls with incurving rims (Type 7), and small flat-based bowls (Types 10-13), primarily associated with eating and drinking activities. The association of Types 3, 5, and 10-13 at Jebel Aruda suggests they formed a drinking and eating set (Van Driel+ and Van Driel-Murray 2023: 210), while a variant of Type 7 (designated Type 8) may have served as lids. Foot-ring variants are occasionally attested, including one documented Type 7 foot-ring vessel at Habuba Kabira-Süd, alongside foot-ring variants of Types 10 and 11 at the same site, and a foot-ring version of Type 4 at Jebel Aruda.

The group also includes vessels designed for specialized functions. These include perforated bowls (Type 6), organic platters and basins (Type 14), deep bowls with central holes (Type 15), fruit stands (Type 16), and oval dishes with egg-cup-like bowls on the rim (Type 17). Types 16 and 17, characterized by delicate designs, likely served luxury or ritual purposes and demonstrate advanced manufacturing techniques. In contrast, Types 6, 14, and 15 are primarily associated with food preparation. Type 6 vessels, likely functioning as ladles, feature central or side holes, possibly for attaching wooden handles. This design differs from Middle Uruk ladles at Sheikh Hassan, where handles are integrated into the base (Boese 1995: 266; see Figure 7), potentially reflecting chronological development or regional stylistic variation. Type 14 platters and basins were likely used for activities such as dough kneading, malting, washing, or slow cooking over indirect heat. Type 15 vessels, possibly colanders, may have been employed in processing fermented foods or dairy products.

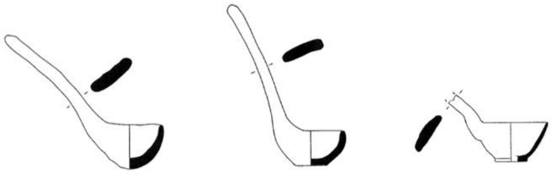


Figure 7: Ceramic Ladles form Sheikh Hassan (After Boese 1995: Figure 9)

3.1.2 Closed vessels

Closed vessels exhibit greater diversity than open forms, encompassing 39 distinct types that are primarily bottles and jars. The most diagnostic Late Uruk shapes in this category are the elongated, narrow vessels (Types 38-41), characterized by their distinctive drooping or straight spouts. Fabricated from porous clay with a coarse sandy temper, these vessels often feature a narrow mouth. Larger versions with drooping spouts and smaller ones with straight spouts may have been intended for different liquids or stages in food manufacturing. However, their service life was likely short due to the porous fabric, which would have promoted rapid evaporation and encouraged bacterial growth (Sürenhagen 1978:99). Together with other spouted forms (Type 34 and Type 37), these vessels are closely connected with liquid pouring and storage, complementing large-scale storage ceramics like Type 45 (torpedo-shaped storage vessels often with shoulder marks), Type 46 (large storage jars) and Type 47 (side rolled vessel), which may have contained grain, beer, water, or oil. The Type 47 was also possibly used for beer brewing. A special form Type 48 (closed oval vessel with narrow spouts) featuring hard, over-fired fabric with minimal permeability, likely used for water transport despite.

In addition to aforementioned storage vessels, the group also includes pots of Types 28/29 and a category of "luxury" wares distinguished by specialized surface treatment and decoration (Types 20-23/26-27). Type 28 comprises globular cooking pots with either rounded or flat, everted rims; their capacity ranges from 2.5 to 7.5 liters. Crafted from common ware, these vessels exhibit blackening from exposure to fire. Some examples of Type 28 are very similar to those of Type 29. Although these two types are grouped together at Habuba Kabira-Süd, they are classified separately at Jebel Aruda. Types 26/27, characterized by a deep red, highly polished fabric, also represent an iconic Uruk form (see Figure 8). At Jebel Aruda, this fabric is termed "Uruk Red", while at Habuba Kabira-Süd, it is described as "red engobed". This fabric is also attested in Types 25/26 and the aforementioned Type 16 fruit stand at both sites. Adorned with lugs, false spouts, and handles, the Uruk Red vessels demonstrate their luxury status, distinguishing them from ordinary vessels made of common fabric. Diatom analysis indicates that some Uruk Red examples may have been imported from southern Mesopotamia (Van Driel+ and Van Driel-Murray 2023: 314).

Other noteworthy forms include the finely made Type 23 bottle, Type 55 gutters and drains, and Type 58 black sherds with white inlay. Type 23, nicknamed "perfume bottles" by the Dutch excavators, may have contained aromatic oils, an interpretation supported by their unique shape and concentration in areas identified as toilets. Type 55 constitutes the sophisticated drainage system at both sites, designed to manage excess rainwater. These were installed in rooms and courtyards at Jebel Aruda and within the road system and buildings at Habuba Kabira-Süd. The distinctive Type 58 sherds, crafted from medium to fine mineral ware, are decorated with a dot-stitch pattern filled with white lime paste. Although these are thought to be imports, their precise origin remains unclear. Their presence at both sites suggests the possibility of a shared trade network by the time.

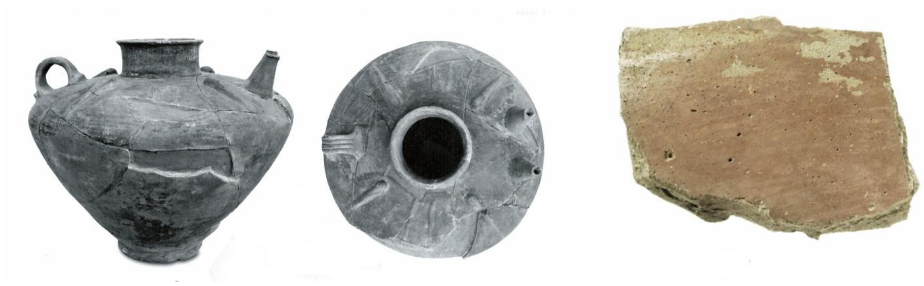


Figure 8: Uruk Red Vessel: Form and Fabric Detail (Edited Based on Strommenger et al. 2014: Plate 4 and Plate 87)

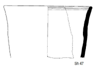
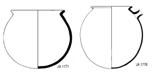
3.2 Unique Forms at Jebel Aruda and Habuba Kabira-Süd

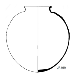
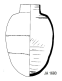
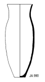
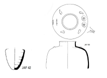
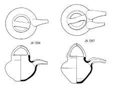
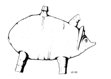
While the ceramic assemblages at Jabel Aruda and Habuba Kabira-Süd share many similarities, each site also produced distinct forms unique to its context. A total of 35 unique forms were identified, with 8 forms (Types 59-66) exclusive to Jabel Aruda and 27 forms (Types 67-93) exclusive to Habuba Kabira-Süd (see Table X). Again, these forms are categorized as open or closed vessels, with their quantities recorded and listed in descending order within each category in Tables 4 and 5.

3.2.1 Unique Forms in Jebel Aruda

The typological analysis indicates that Jebel Aruda yielded eight distinctive pottery forms, including one open form and seven closed forms, encompassing globular pots, bottles, and animal figurines. These vessels are characterized by their unique fabric, morphology, or a combination thereof, offering insights into the diversity of the site's ceramic assemblage.

Table 4: Pottery Categories Exclusive to Jabel Aruda

	Type	Categories	Description	Pottery Illustration (not to scale)	Quantity
Open Vessel	59	Cat. 43	Massive basin		3
Closed Vessels	60	Cat. 16	Calcite tempered cooking pots		19

	61	Cat. 29	Global pots [8]		13
	62	Cat. 19	Round base vessels with straight and high neck made of chaff tempered fabric		11
	63	Cat. 24 C	Open bottles with narrow base but no shoulder		10
	64	Cat. 13 (Part of)	A group of 'Uruk Red copies' made of yellowish soapy fabric		6
	65	Cat. 42	"Teapots" with basket handle		4
	66	Cat. 43	Hedgehog shaped vessel		3

3.2.1.1 Open vessels

Type 59, the sole open form unique to Jebel Aruda, is represented by three examples. One specimen, registered as Sh 47 and constructed using the slab-building technique, is remarkable for its exceptional size, measuring 1.55 meters in height and 1.68 meters in diameter. Its scale is particularly striking when compared with the second-largest example, which has a preserved height of 0.56 meters and a considerably smaller rim diameter. Although basins are common at both sites, the extraordinary dimensions of Sh 47 raise questions about its specific function and contextual use. Its size suggests that it may have served purposes beyond ordinary storage, potentially in ceremonial or communal contexts, or in association with particular social groups. Nevertheless, as this represents a single specimen, the evidence cannot be regarded as statistically significant, and the possibility that similarly large basins once existed at Habuba Kabira-Süd but were not preserved cannot be excluded.

3.2.1.2 Closed vessels

Among the closed forms, Types 60, 61, and 64 are distinguished by their unique fabrics. Type 60, with a total of 19 examples, is the most prevalent of these forms at Jebel Aruda. With blackened and burnt marks, its cooking function is conclusive. These vessels were crafted using a hard paste with heat-resistant calcite temper, indicating they were higher-quality cooking pots compared to the standard Type 28 (Cat. 15/FG 22.4). The fabric has no parallel within other ceramic categories at the site. Three large and heavy examples of this type have spout-like handles, possibly serving for milk or yogurt processing; a stick could be inserted into the handle to pour the contents out. The relatively high frequency of Type 60 vessels signifies their important role in the daily subsistence activities at Jebel Aruda, potentially reflecting culinary practices not seen at Habuba Kabira-Süd, where this pottery type is absent. Type 61, known as "blister pots," is the second most frequent unique form, with 14 specimens. Constructed from a non-local, thin, blistered fabric, these vessels are often over-fired, resulting in an extremely hard texture and greenish or grey coloration. Its impermeability suggests it is well-suited for water storage, though its exact function is uncertain. Type 64, termed "Uruk red copies," exhibits formal and decorative similarities to Type 26 Uruk red vessels (Cat. 14/FG 31) but is manufactured from a buff fabric with a soapy self-slip. Represented by only six specimens, these imitations are less statistically significant compared to Types 60 and 61.

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[8] Fabric differs between the sites: Jebel Aruda utilized non-local blister fabric while Habuba Kabira-Süd relied on local clay sources

Three closed forms, Types 63, 65, and 66, are notable for their unique morphologies at Jebel Aruda. Type 63 comprises high, narrow vessels with a wide mouth and no shoulder. Their form suggests possible functions such as beer production or storage, holding decanted liquids, or containing preserved foods that could be easily scooped. Type 65, small squash pots with a basket handle nicknamed "teapots," are among the most delicate and rare forms at Jebel Aruda (see Figure 9, lower right). They are clearly associated with drinking functions, though examples featuring two spouts are unusual. Their basket handle may indicate they were modeled after metal prototypes. Type 66, the "hedgehog" figurines, represents the most distinctive pottery form at Jebel Aruda in terms of shape. While bovine-shaped figurines are common during the Uruk period, the hedgehog is uniquely favored at Jebel Aruda. The pot depicts details of a hedgehog, including eyes, ears, a snout, legs, spines, and a filling and pouring spout on the back, indicating a highly skillful pottery technique (see Figure 9, left and upper right). All three examples (JA 34, JA 140 and JA 92) are excavated in the complex SI. The house is interpreted by excavator as a central hub for administration, storage and distribution due to its strategic location and rich contents. The presence of hedgehog figurines may reflect a unique cultural preference or a connection to temple activities at Jebel Aruda.

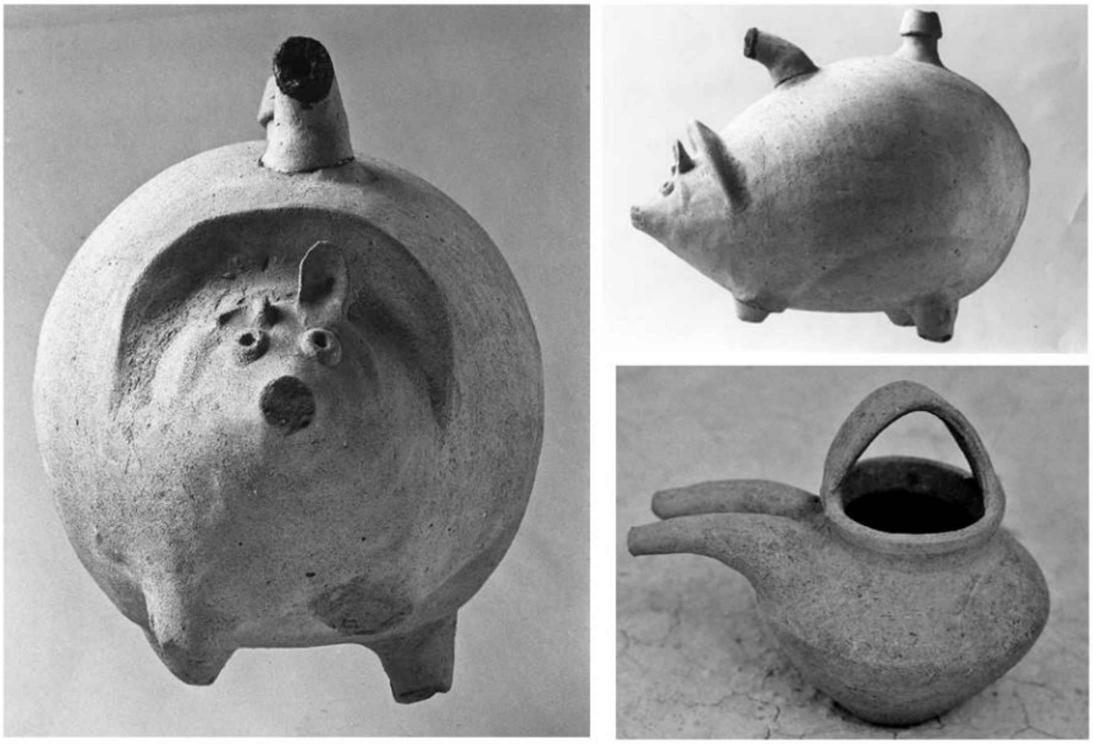


Figure 9: Type 66 Hedgehog Figurines (left and upper right) and Type 65 Double Spouted "Teapots" (lower right) from Jebel Aruda (Edited Based on Van Driel+ and Van Driel-Murray 2023: Figure 5.3 and Figure 7.4)

Type 62, characterized by its straight neck and thick base, is among the most characteristic vessels in the Jebel Aruda assemblage due to its distinctive fabric and morphology. The pot appears to have been coated with a whitish slip and wiped with grass on the exterior. With a capacity of 12.2 liters, twice that of the larger Type 28 pots, this vessel likely served as a cooking pot, though its narrow neck makes scooping out contents difficult. Its chaff-tempered fabric, often exhibiting a dark core due to incomplete oxidation and reddish interior and exterior surfaces, corresponds to the "Chaff-Faced Simple Ware" (CFSW) identified by Braidwood and Braidwood (1960: 232-238) in the Amuq Plain's Phase F pottery. This fabric is also attested at other Late Chalcolithic sites, such as Arslantepe Phase VII in southeast Anatolia, Hacinebi Phase A/B1 in the Middle Euphrates Valley, and Hammam et-Turkman Phase V in the Balikh region (Marro 2010:37-39). Compared to mineral-tempered ceramics, organic-tempered fabrics offer advantages such as shorter firing times, lower firing temperatures, and greater resistance to thermal shock (Akkermans 1988: 292). In terms of form, Type 62 closely parallels vessels from

Hammam et-Turkman VA (see Figure 10), whose chronology may be earlier than that of Amuq F (Akkermans 1988: 321). Several other high-collared forms, morphologically similar to Type 63, are also found in Amuq F, reinforcing its close connection to local Chalcolithic traditions. The presence of 11 examples of this type at Jebel Aruda is therefore unlikely to be accidental and its uniqueness is further evidenced by its spatial and contextual distribution, which will be discussed in Chapter 5.

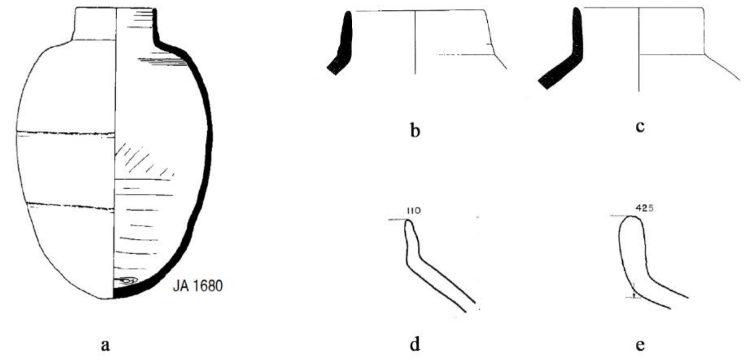
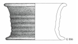
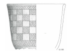
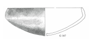
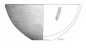
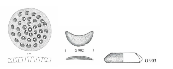
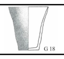
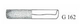
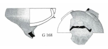
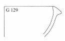
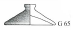
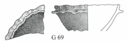
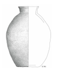
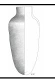
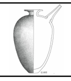
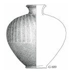
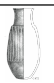
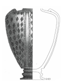
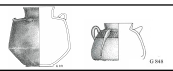
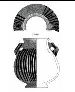
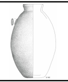
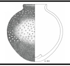
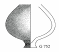


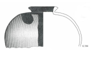
Figure 10: Type 62 Pot from Jebel Aruda (a) and Parallel Vessels from Hammam et-Turkman (b-c) and the Amuq Plains (d-e)

3.2.1 Unique Forms in Habuba Kabira-Süd

The assemblage from Habuba Kabira-Süd (see Table 5) includes 11 unique open vessels (Types 67-77) and 16 unique closed vessels (Types 78-93), encompassing a wide range of shapes and functions. These comprise bowls (Types 68-70, 72, 74-75, 77), jars (Types 89, 90), pots (Types 84, 87, 91, 93), bottles (Types 78-82, 86, 90), stands (Type 67), vessels that may have served as lids (Types 73, 76), tools (Type 71), and painted or decorated vases (Types 83, 85). While it is not feasible to describe each type in detail as was done for Jebel Aruda, this section focuses on three representative examples: the pottery stand (Type 67), the flat-bottomed pot (Type 78), and the elongated bottle (Type 82). The first two are the most statistically significant, occurring in the greatest quantities within their respective categories, while the last is distinguished by its unique decorative treatment.

Table 5: Pottery Categories Exclusive to Habuba Kabira-Süd

	Type	Group (HK)	Registration Number	Description	Pottery Illustration (not to scale)	Quantity
Open Vessels	67	FG 47	G 886-897	Pottery stands		13
	68	FG 10.1	G 101-110	Round bottomed, steep-walled bowls		12
	69	FG12.2	G 165-167	Bent-walled lid (bowl?) with centrally perforated bases		7
	70	FG11.1	G 118-122	Semi-soherical bowls with flat, swept rim		5
	71	FG 50/51/52	G 901-904	Tools or installations: Perforated disc, biconic rings and lunate-shaped scraper		5
	72	FG3.1	G16-18	Steep-walled bowls		4
	73	FG12.1 (Part of)	G 147/162-164	Straight wall flat base lid (bowl?)		4
	74	FG12.3	G 168-169	Bent-walled bowls with ear handles on the edge		2
	75	FG11.3	G 128-129	Hammerhead rim bowl		2
	76	FG 5	G 65	Lid?		1
	77	FG 7 (G 69)	G 69	Double rim bowl		1
Closed Vessels	78	FG 59 (Part of)	G 961-978	Flat-bottomed pots with a cylindrical neck and projecting beaded rim (chaff tempered)		22
	79	FG 63	G 987-1000	Round-bottomed bottles with a cylindrical neck (chaff tempered)		14
	80	FG 38	G 695-702	Flat-bottomed bottles with a long, straight spout		11
	81	FG 37.2	G 689-694	Bottles with a narrow cylindrical neck and ring base, some with pseudo-reserved slip		8
	82	FG 45 (Part of)	G 866-873	Elongated bottles, some with pseudo-reserved slip		8
	83	FG 42	G 823-825	Vases on a profiled base, with vertical ribs and diamond-stamped decoration.		5
	84	FG 44 (Part of)	G 848/850/851	New forms of Type 14 (“inkwell”) attested only at HK		4
	85	FG 64	G 1001-1004	Black-brown painted vase with cord loops and ring base		4
	86	FG 61	G 982-984	Flat-bottomed, bottle-like vessels with a funnel rim		3
	87	FG 22.2 (G 303)	G 303	Round pots full of holes		2
	88	FG40 (Part of)	G 752-753	Special bottles with narrow high stand		2

	89	FG 22.5 (G 384)	G 384	Huge rounded jar with bitumen/paiting		1
	90	FG 28 (G 532)	G 532	Large flat-bottomed jar with asphalt painting		1
	91	FG 22.2 (G 279)	G 279	Globular pots with drooping spout and incised decoration		1
	92	FG 27 (G 485)	G 485	Unique four-eyelet vessels		1
	93	FG 67 (G1005)	G1005	Spherical funnel-rim pot with blackish spots on the exterior		1

Pottery stands (Type 67) are the most prominent open shape unique to Habuba Kabira-Süd. This stands in sharp contrast to Jebel Aruda, where objects serving a stand function are less common and were primarily made from non-clay materials like stone. One example from Jebel Aruda is simply a re-used, inverted pottery rim, rather than a purpose-built stand. Only five clay stands were identified there, but their supporting function remains unclear. Even if they were used as stands, their shapes differ significantly from those at Habuba Kabira-Süd (see Figure 11). At Habuba Kabira-Süd, however, pottery stands are both abundant and typologically diverse. The 13 recorded examples range from short to tall stands, some with straight openings and others with flaring rims, designed to support vessels of different shapes. Most of the examples were wheel-made using standard Fabrics 1 and 2 and several specimens also feature incised decorations. The largest example, G891 (Figure 11: e), is exceptional for a height of 29.5 cm and a red engobed exterior. Its unusual scale and refined surface treatment suggest it may have been part of a luxury pottery set or used to display prestige vessels in a ritual or high-status context. In addition, three hand-formed stands were also identified: two in Fabric 18 (G897, G897.1) and one in Fabric 19 (G896). Fabric 18 is tempered with minimal chaff and was almost exclusively used for BRBs, while Fabric 19 contains 30-50% medium-fine to coarse chaff temper and was predominantly used for BTs. The variety of fabrics used for this single form is particularly noteworthy and not commonly seen on other types.

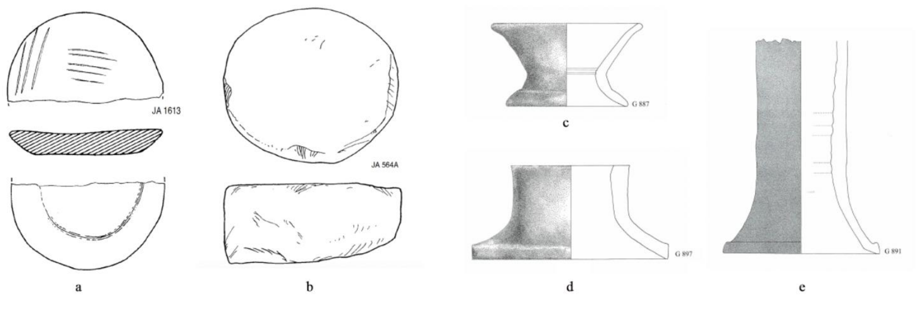


Figure 11: Stands from Jebel Aruda (a-b) and Habuba Kabira-Süd (c-e) (not to scale)

Within the closed form group, the Type 78 flat-bottomed pots are the most frequent unique form. Belonging to the "private production" Group IV, this type represents the best example of Amuq F chaff-faced ware in the Habuba Kabira-Süd corpus. These bottles share the same black-core fabric as the Type 62 straight-necked pots found at Jebel Aruda, and closely resemble forms attested in Amuq Phase G (see Figure 12). A broader examination, however, demonstrates that Type 78 is not the only vessel linked to non-Uruk traditions. Other examples include the chaff-tempered bottles Type 79 and Type 86, which are also in Group IV and exhibit comparable characteristics. Furthermore, Type 93 pots, distinguished by blackened surface spots, may have originated in southeastern or central Anatolia. Additional links to local traditions are evident in vessel morphology. For instance, the Type 75 "hammer-head" bowls parallel forms documented in the pre-contact Phases A and B1 at Hacinebi (Pollock and Coursey 1995: Fig. 5A), whereas the Type 77 double-rim jars correspond to examples from Tepe Gawra Level XI (Tobler 1950: Plate CXLVI: 405-407). Although made of grit-tempered material typical of Uruk traditions, their close morphological connection to indigenous cultures suggests possible imitation or influence. This overall pattern indicates a significant impact of local Chalcolithic traditions on the pottery assemblage at Habuba Kabira-Süd.

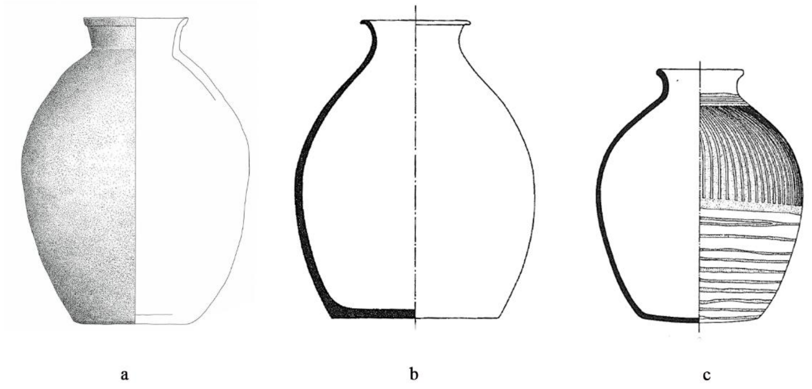


Figure 12: Type 62 from Habuba Kabira-Süd (a) and Amuq F Plain Simple Ware (b) and Reserved-Slip Ware (c) (not to scale)

Another distinctive closed form, the Type 82 elongate bottle, is unique to Habuba Kabira Süd for its shape and "pseudo-reserved slip" decoration (see Figure 13), absent at Jebel Aruda. This type has a slim body, a wide mouth at least as broad as the base, a long neck, and a narrow shoulder. The tallest example (G872) measures 28.2 cm in height, with a rim diameter of 7.4 cm, a base diameter of 5.8 cm, and a maximum diameter of 12.8 cm. Unlike true reserved slip, in which the surface was first engobed and then strips were erased diagonally, pseudo-reserved slip was produced by diagonally smoothing the wet surface of the vessel. After firing, these smoothed strips could be wiped out by hand (Strommenger et al. 2014:151). This decorative technique also appears on other unique forms, such as Type 81, and several common forms, including Type 42. At Habuba Kabira-Süd, 791 (7.57% of 10,617 surface-treated sherds) exhibit pseudo-reserved slip, with the decoration most prevalent on Type 82, where it adorns 37.5% of examples (Strommenger et al. 2014:152) At Jebel Aruda, true reserved slip was recorded, but pseudo-reserved slip is not mentioned in the excavation report. The prevalence of this decorative treatment at Habuba Kabira-Süd, contrasted with its absence at Jebel Aruda, is notable and may point to differences in technical traditions or aesthetic preferences between the two sites.

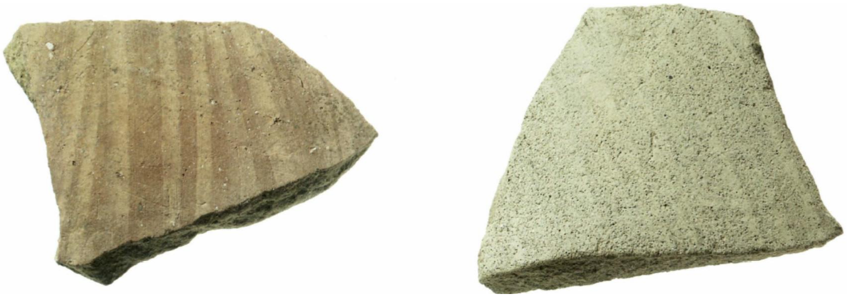


Figure 13: True Reserved Slip (left) and Pseudo Reserved Slip (right) (Edited Based on Strommenger et al. 2014: Plate 4)

In conclusion, the analysis of ceramic forms at Jebel Aruda and Habuba Kabira Süd yields three key observations: (1) while both sites share a majority of vessel forms, each produced unique forms, with Habuba Kabira-Süd exhibiting a larger and more diverse corpus of unique forms compared to Jebel Aruda; (2) unique forms at both sites demonstrate connections to local Chalcolithic traditions, though Habuba Kabira Süd displays a greater number of such forms; and (3) the “pseudo-reserved slip” surface treatment, prevalent and diagnostically significant exclusively at Habuba Kabira-Süd, indicates distinct pottery decoration techniques or aesthetic preferences not observed at Jebel Aruda.

4. Frequency Comparison

This section analyzes the frequency of pottery assemblages at Habuba Kabira Süd and Jebel Aruda. The comparison employs a three-part analytical framework. First, all vessel forms are subjected to a preliminary analysis in order to provide a broad overview of each assemblage. Second, a focused frequency analysis of shared forms highlights similarities and differences in their relative prevalence. Third, a separate frequency analysis of local Chalcolithic-related pottery investigates non-Uruk elements, which constitute a substantial proportion of the unique forms at both sites. Unidentifiable sherds, as well as BRBs and BTs, were excluded from all analyses. Frequency data were sourced from published inventory lists.

Direct comparison of absolute counts between the two sites was not feasible due to differing registration methods: Jebel Aruda's Dutch team used a vessel-oriented system, while Habuba Kabira-Süd's German team employed a sherd-oriented system. To address this, vessel frequencies at Habuba Kabira-Süd were manually reconstructed from the inventory lists, yielding 1,291 vessels for the first two analyses. Despite minor discrepancies exist between pottery descriptions and inventory entries in the Habuba Kabira-Süd's report, the inventory data were applied consistently to preserve comparability with Jebel Aruda. For the third analysis, which focuses on chaff-tempered fabrics, the original sherd-based data from Habuba Kabira-Süd were retained, as conversion to vessel counts would have introduced considerable inaccuracies.

4.1 Preliminary Comparison of Ceramic Forms from Jebel Aruda and Habuba Kabira-Süd

A preliminary frequency analysis was conducted on 1,330 examples from Jebel Aruda and 1,291 vessels from Habuba Kabira-Süd to assess the relative distribution of pottery forms at the two sites. This comparison provides an overall profile of each assemblage and indicates whether common forms predominate. In cases where they do, subsequent analyses of shared forms can be undertaken with greater validity. Where unique forms occur in substantial numbers, however, they may exert a significant influence on assemblage composition and necessitate a broader interpretive framework.

The results (see Figure 14) show that, following BRB and BT, Cat. 3 goblets (Type 3), Cat. 4 bowls (Type 4), and Cat.15 pots (Type 28) are the most frequent forms at Jebel Aruda, while rare types such as Cat. 38 pots (Type 48), Cat. 43 hedgehog vessels (Type 66), and Cat. 40 dishes (Type 17) occur only in very small numbers. The distribution is highly uneven, with the fifteen most common categories comprising over 70.30% of the total assemblage (935 out of 1,330). At Habuba Kabira-Süd, the most common forms are FG 27 (Type 22), FG 31 (Type 26), FG 41 (Type 35) and FG 22.4 (Type 28/29). At the other end of the spectrum, foms such as FG 50 (Type 71), FG 66 (Type 51) and FG 67 (Type 58) are represented by only a single example each. Although the distribution here is less heavily skewed toward the dominant types than at Jebel Aruda, the fifteen most common forms still constitute 48.72% of the total assemblage (629 out of 1,291).

This preliminary analysis confirms that none of the unique forms discussed in the preceding section 3.2 occur in substantial numbers comparing common forms. Even the most frequent examples, Cat. 16 (Type 60) at Jebel Aruda and FG 59 (Type 78) at Habuba Kabira-Süd, do not rank among the top twenty types. The observation supports the rationale for analyzing common forms separately, as they provide a more reliable indication of principal distributional trends at the two sites. By contrast, unique forms are quantitatively marginal and exert little influence on the overall conclusions.

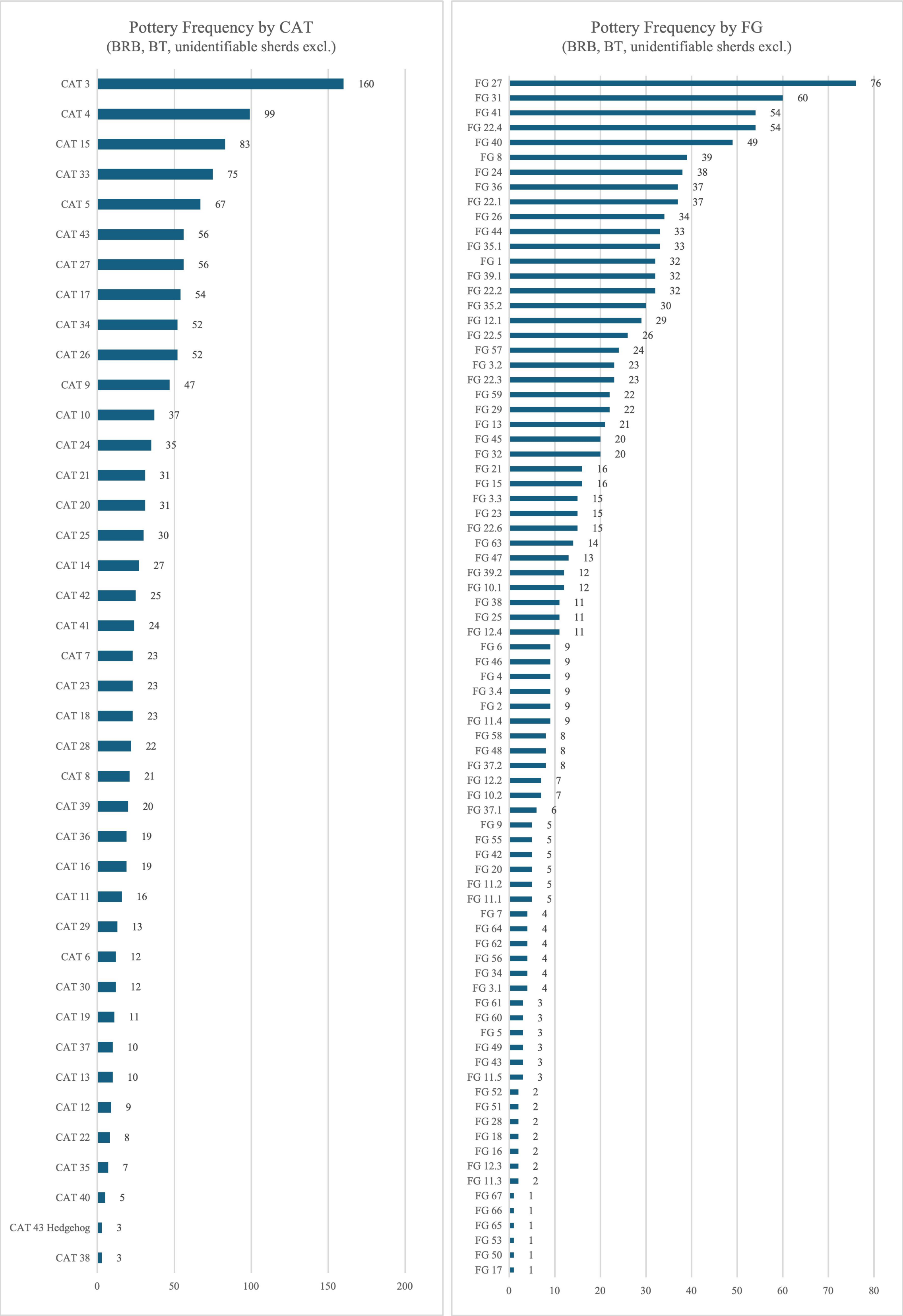


Figure 14: Pottery Frequency at Jebel Aruda (left) and Habuba Kabira-Süd (right)

4.2 Frequency Comparison of Common Forms Shared at Jebel Aruda and Habuba Kabira-Süd

Following the preliminary analysis, a frequency analysis of common forms was conducted with a clear rationale. Variations in typological outcomes between Habuba Kabira-Süd and Jebel Aruda are evident in the Chapter 3, where certain forms at Jebel Aruda correspond to subgroups at Habuba Kabira-Süd, while some Habuba Kabira-Süd forms align with multiple Jebel Aruda categories. To facilitate a meaningful quantitative comparison, adjustments to the typological groupings were necessary. Consequently, 34 comparison groups (Comp 1 to Comp 34) were established based on observed correspondences.

The table below outlines the composition of these groups and their relationships across the two sites (see Table 6). The groupings were constructed according to two principles designed to maximize comparability: (1) objects within each group must display clear morphological parallels, and (2) combinations and exclusions were carried out with minimal intervention, preserving the original classifications at each site as far as possible and minimizing the need for extensive recounting.

For example, Comparison Group 17 (Comp 17) includes Types 28 and 29, corresponding to Categories 15 and 17 at Jebel Aruda and align with FG 22.3 and FG 22.4 at Habuba Kabira-Süd. This adjustment is warranted given the morphological similarity between Types 28 and 29. At Jebel Aruda, they are assigned to separate categories (Cat. 15 for smaller, globular vessels and Cat. 17 for egg-shaped forms). However, the excavator notes that many vessels are difficult to differentiate, resulting in a potential overrepresentation of Cat. 15 (Van Driel+ and Van Driel-Murray 2023: 222). At Habuba Kabira-Süd, these forms are divided into FG 22.3 and FG 22.4, primarily based on rim characteristics rather than overall shape. Three spouted variants assigned to FG 22.4 are morphologically distinct and were therefore excluded from Comp 17. Similarly, the combination of Types 22 and 42 in Comparison Group 13 (Comp 13) is justified. While these forms are separated at Jebel Aruda into Cat. 10 C/D and Cat. 27 on the basis of size and the presence of features such as base rings, their strong morphological similarity supports their treatment as a single group.

Table 6: Comparison Groups and Pottery Compositions at Jebel Aruda (JA) and Habuba Kabira-Süd (HK)

(1) Comp	(2) Type	(3) JA Category	(4) HK Group	(5) JA Quantiy	(6) HK Quantity	(7) JA Rank	(8) HK Rank	(9)=(7)-(8) Rank Difference
1	3	Cat. 3	FG 1	160	32	1	13	-12
2	4	Cat. 4	FG 8	99	39	3	5	-2
3	5	Cat. 5	FG 3.2	67	23	5	16	-11
4	6	Cat. 6	FG 6	12	9	23	24	-1
5	7+8	Cat. 7	FG 11.2 + FG 12.1	23	34	13	8	5
6	9	Cat. 8	FG 22.6	21	15	16	20	-4
7	10	Cat. 9A	FG 3.4	7	9	27	25	2
8	11	Cat. 9B	FG 3.3	15	15	22	21	1
9	12	Cat. 9C	FG 10.2	30	7	10	29	-19
10	13	Cat. 9D	FG 5+ 1 example from FG 7	6	4	30	31	-1
11	20	Cat. 10A	FG 44	4	33	32	10	22
12	21	Cat. 10B	FG 25	7	11*	28	22	6
13	22+ 42	Cat. 10C/10D + 16 examples from Cat. 27	FG 27	42	76	7	1	6
14	23	Cat. 11	FG 24	16	38	21	6	15
15	26	Cat. 14	FG 31	27	60	12	3	9
16	27	2 examples from Cat. 28	FG 29	2	22	34	17	17
17	28+29	Cat. 15 + Cat. 17	FG 22.3+ FG 22.4 - 3 spouted versions not belonging to this group	137	74	2	2	0
18	30	Cat. 18	FG 36 (G 661)	23	1	14	32	-18
19	31	Cat. 20	FG 57	31	24	9	15	-6
20	33	Cat. 22	FG 37.1	8	6	26	30	-4
21	34	Cat. 23	FG 22.3 (G305)	23	1	15	33	-18
22	35	Cat. 24A	FG 41	20	54	17	4	13
23	36+45	Cat. 24B+ Cat. 33	FG 36 -1 example not belonging to the group	80	36	4	7	-3
24	37	Cat. 25	FG 39.2+FG 32	30	32	11	14	-3
25	38+39+40+41	Cat. 26	FG 39.1 + 1 piece from FG 45	52	33	6	11	-5
26	42	Cat. 27- 16 examples not belonging to this group	FG 26	40	34	8	9	-1
27	27	Cat. 28 -2 examples assigned to Comp 16	FG 35.1	20	33	18	12	6
28	44	Cat. 30	FG 58	12	8	24	28	-4
29	47	Cat. 35	FG 22.5 (G381)	7	1	29	34	-5
30	14	Cat. 36	FG 15	19	16	20	19	1
31	15	Cat. 37	FG 11.4	10	9	25	26	-1
32	48	Cat. 38	FG 46	3	9	33	27	6
33	16	Cat. 39	FG 12.4	20	11	19	23	-4
34	17	Cat. 40	FG 13	5	21	31	18	13

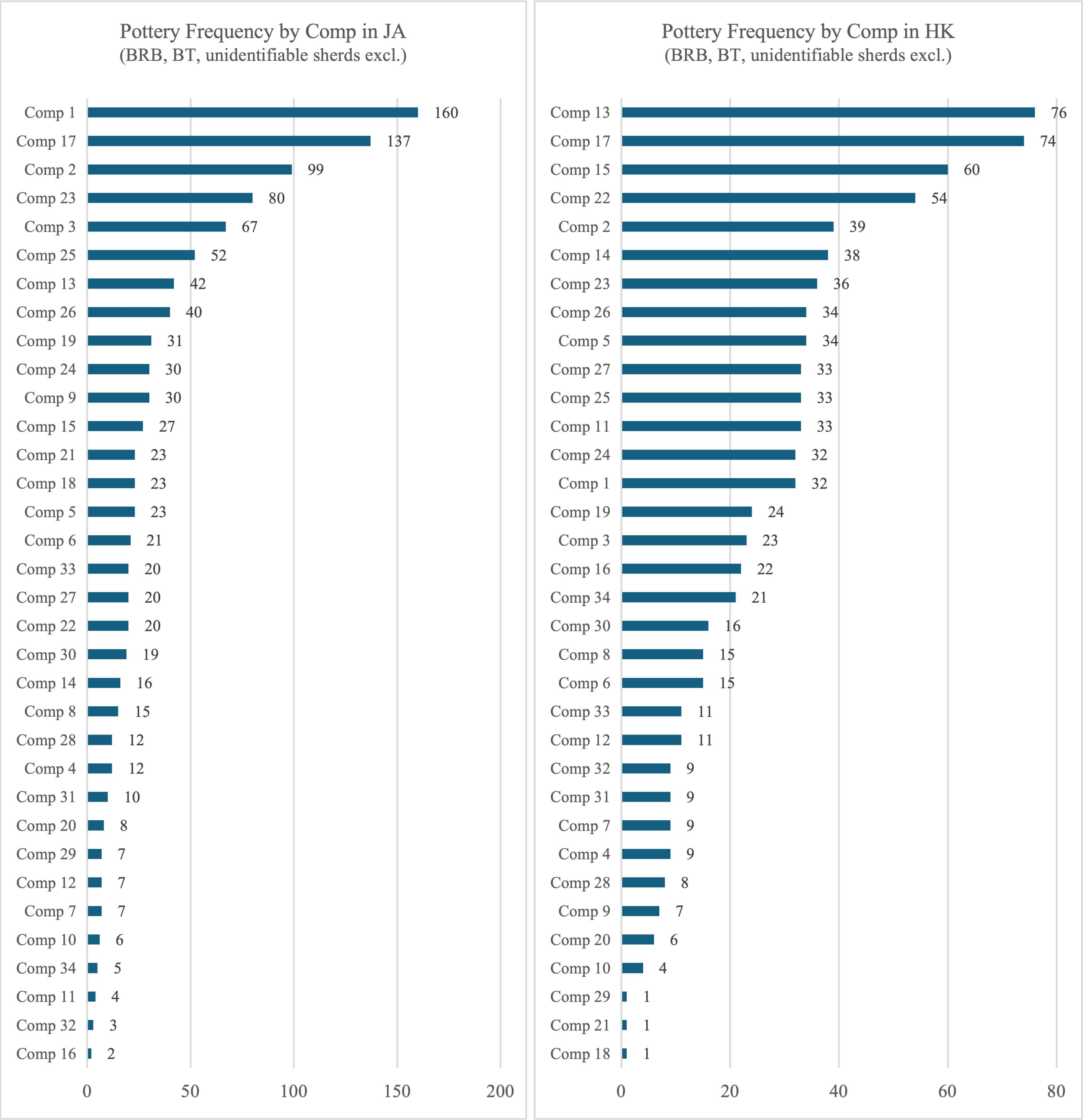
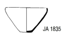
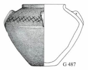
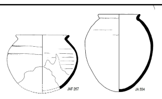
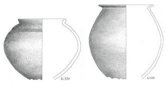
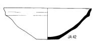
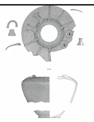
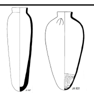
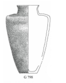
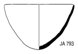
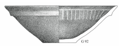


Figure 15: Comparable Pottery Frequency at Jebel Aruda (JA) and Habuba Kabira-Süd (HK)

The frequency analysis of the adjusted pottery comparison groups reveals distinct patterns at Jebel Aruda and Habuba Kabira-Süd (see Figure 15 and Table 7). At Jebel Aruda, the most prevalent forms are Comp 1, Comp 17, Comp 2, Comp 23, and Comp 3, corresponding to Type 3 simple goblets, Types 28/29 globular and egg-shaped pots, Type 4 beaked-rim bowls, Types 36/45 torpedo-shaped and high narrow storage vessels, and Type 5 small round-based bowls, respectively. In contrast, at Habuba Kabira-Süd, Comp 13, comprising Types 22/42 small and large pots with lugs and incised decoration, ranks highest, followed by Comp 17 (Type 28/29 pots, also the second-ranked group at Jebel Aruda), Comp 15 (Type 26 Uruk red vessels), Comp 22 (Type 35 kink-shouldered flat-bottomed vases), and Comp 2 (Type 4 bowls, also the third-ranked group at Jebel Aruda).

Table 7: Top Five Most Frequent Comparison Groups at Jebel Aruda and Habuba Kabira-Süd

Jebel Aruda				Habuba Kabira-Süd			
Rank	Comp (Type)	Illustrations	Quantity	Rank	Comp (Type)	Illustrations	Quantity
1	Comp 1 (Type 3)		160	1	Comp 13 (Type 22/42)		76
2	Comp 17 (Type 28/29)		137	2	Comp 17 (Type 28/29)		74
3	Comp 2 (Type 4)		99	3	Comp 15 (Type 26)		60
4	Comp 23 (Type 36/45)		80	4	Comp 22 (Type 35)		54
5	Comp 3 (Type 5)		67	5	Comp 2 (Type 4)		39

The results show that the top five forms at the two sites overlap in two cases but diverge in three. The shared forms relate to cooking (Comp 17) and eating/drinking (Comp 2), pointing to broadly similar practices in daily activities. The three non-overlapping forms, however, highlight marked contrasts. At Jebel Aruda, these are drinking cups, storage vessels, and round bowls, all tied to routine, practical functions. At Habuba Kabira-Süd, by contrast, they consist of two luxury types (lugged vessels and Uruk red ware) and a storage vessel likely used for valuable liquids (kink-shouldered bottle). This pattern runs counter to expectations. Jebel Aruda is generally interpreted as a religious center, which should imply a greater presence of prestige or ritual ceramics. Yet it is Habuba Kabira-Süd that produces higher frequencies of luxury pottery. Should this prompt a reconsideration of the established interpretations of site function? Alternatively, if Jebel Aruda's interpretation remains valid, does the contrast point to differences in status, with Habuba Kabira-Süd holding a higher rank in the settlement hierarchy? Or could the divergence instead reflect variations in the composition of the resident populations at the two sites?

Additionally, the frequency rankings of comparable forms exhibit substantial variations across several groups (see Table 6: Columns 7-9). The most notable cases are Comp 11, which exhibits the greatest disparity of 22 ranks across the entire corpus, and Comp 1, which shows the largest difference of 12 ranks within the top five forms previously discussed. The so-called "inkwell" vessels of Comp 11, whose precise function remains uncertain, not only occur in greater numbers at Habuba Kabira-Süd than at Jebel Aruda but also display a wider morphological range. Their fine fabrics, incised decoration, and lugs mark them as luxury ceramics. The causes of this divergence merit closer consideration. Does the higher frequency and morphological diversity at Habuba Kabira-Süd reflect stronger local demand and preference? Or might it be connected to Habuba Kabira-Süd's role as a trade hub, where increased commercial activity encouraged the circulation and accumulation of diverse ceramic types? In contrast, Comp 1, comprising small goblets, constitutes the most prevalent type at Jebel Aruda, with 160 examples. This count far surpasses the second most abundant form, Comp 17, by 23 vessels, dominating the assemblage by frequency. These goblets are hypothesized to have functioned as "teacups" for drinking or eating, possibly employed in nested sets of small, medium, and large sizes (see Figure 16). At Habuba Kabira-Süd, however, Comp 1 ranks only 13th, with a total of 32 pieces, indicating a comparatively minor role. This marked discrepancy raises questions about whether it reflects differing cultural practices between the two sites, a topic that remains open for further discussion.

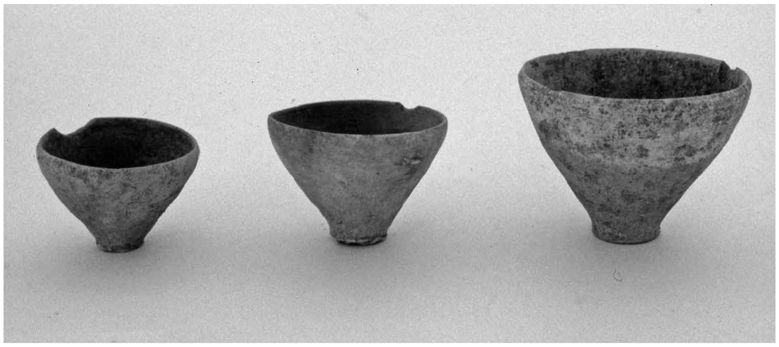


Figure 16: Goblets of Different Sizes from Jebel Aruda (After Van Driel+ and Van Driel-Murray 2023: Figure 8.10)

In summary, the frequency analysis of common forms demonstrates that differences in vessel quantities outweigh similarities between the two sites. Apart from the Comp. 17 cooking pots and Comp. 2 bowls, which show comparable frequency ranks, most other forms display marked variation. In the top five categories, Jebel Aruda is dominated by the teacup (Comp 1) and other vessels associated with daily activities, whereas Habuba Kabira-Süd is characterized by a greater presence of ritual and status-related forms. Beyond the top five, additional types such as the inkwell also exhibit considerable differences in rank, further highlighting the distinct composition of the two assemblages.

4.3 Frequency Comparison of Local Late Chalcolithic Related Forms

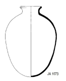
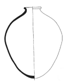
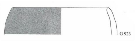
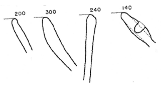
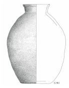
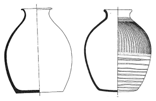
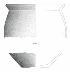
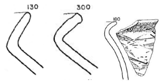
With the frequency analysis of unique pottery forms mentioned in Section 3.2, this section shifts focus to local late chalcolithic related forms, which constitute a significant portion of these unique forms. Although Habuba Kabira-Süd and Jebel Aruda are commonly regarded as Uruk enclaves, they are not exclusively "Uruk" in character. Alongside the numerous typical Uruk types, both sites contain a substantial component of local Chalcolithic elements, recognizable through similarities in fabric, shape, or both.

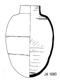
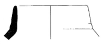
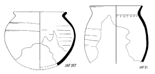
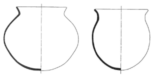
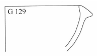
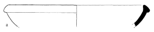
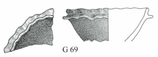
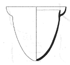
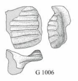
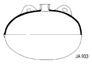
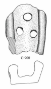
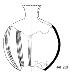
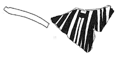
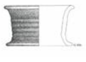
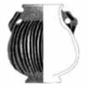
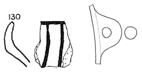
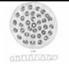
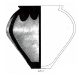
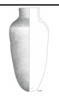
The occurrence of Amuq F and Amuq G-related fabrics has already been documented and assigned to the "private production" Group IV in the Habuba Kabira-Süd excavation report. This group comprises objects made of wares corresponding to the "chaff-faced simple ware," "red double-slipped ware," "coarse cooking-pot ware," and "well-made cooking-pot ware" of the Amuq F phase, as well as the "first cooking-pot ware" and "second cooking-pot ware" from the early Amuq G phase (Strommenger et al. 2014:154). A closer examination of Jebel Aruda, however, demonstrates that local late Chalcolithic-related fabrics are also well represented there, even though the associated forms were not formally segregated into a distinct group. In addition to the BRBs, BTs, and platters, which are common at Uruk-related sites, organic or chaff-tempered fabrics occur in at least 8 further categories. These include the purely chaff-tempered Cat. 19 (Type 62) and Cat. 30 (Type 44); heavily organic-tempered examples in Cat. 33 (Type 45) and Cat. 20 (Type 31); medium to lightly organic-tempered examples in cooking pot Cat. 15 (Type 28) and large storage jar Cat. 34 (Type 46); a single vessel in Cat. 38 (Type 48); and several examples in Cat. 43 (Types 55/57).

In addition to fabric, morphological traits characteristic of indigenous traditions are also evident at both sites. At Habuba Kabira-Süd, these include aforementioned hammer-head bowls (Type 75), double-rim jars (Type 77), and flat-bottomed bottles (Type 78), which are absent from Jebel Aruda. Shared forms, however, include hole-mouth jars (Type 56), baggy pots (Type 31), and cooking pots (Type 28), all of which find parallels in Amuq F or G assemblages. Notably, Type 28 cooking pots display considerable variability in shape, yet most variants find close parallels in the "coarse cooking-pot ware" and "well made cooking-pot ware" of Amuq F (Braidwood and Braidwood 1960 :235), with occasional examples also exhibiting organic temper.

Based on an analysis of fabric and morphology, this study identifies three groups of local Chalcolithic-related vessels (see Table 8): (A) purely local forms that exhibit both the fabrics and shapes characteristic of local traditions; (B) hybrid forms that share either the fabric or the shape of local traditions; and (C) ambiguous chaff-tempered pottery that does not fit into groups (A) or (B), and whose cultural affiliation (Uruk or local) remains uncertain. This classification provides a framework for subsequent frequency analysis to quantify the non-uruk element within the assemblage. To this end, an assessment requires calculating the combined proportions of Groups A, B, and C.

Table 8: Local Late Chalcolithic-Related Vessels at Jebel Aruda and Habuba Kabira-Süd

A. Purely Local Forms				
Description	Pottery Type (JA/HK)	Pottery Illustrations	Representative Local Chalcolithic Assemblage	Pottery Illustrations
Chaff tempered fabric and local shapes	Type 31 (Cat. 20/ FG 57)		Amuq G (Braidwood and Braidwood 1960 - Fig 203:8)	
	Type 56 (Cat. 43/FG 56)		Amuq F (Braidwood and Braidwood 1960 -Fig 176:1-4)	
	Type 78 (-/ FG 59 and 61)		Amuq G (Braidwood and Braidwood 1960 - Fig 203:11; Fig 219:2)	
	Type 44 (Cat. 30/ FG 58)		Amuq F (Braidwood and Braidwood 1960 - Fig 176:18/ 25; Fig 178: 8)	

	Type 62 (Cat. 19/ -)		Hammam VA (Akkermans, Peter M. M. G 1988 - Plate 102:52)	
B. Hybrid Forms				
Description	Pottery Type (JA/HK)	Pottery Illustrations	Representative Local Chalcolithic Assemblage	Pottery Illustrations
a. Local shapes but non-chaff-tempered fabric	Type 28/29 (Cat. 15 and 17/ FG22.3-22.4)		Amuq F/G (Braidwood and Braidwood 1960- Fig 175:2-4)	
	Type 75 (-/FG 11.3)		Hacinebi A and B1 (Pollock and Coursey 1995: Fig. 5A)	
	Type 77 (-/ FG 7)		Gawra Level XI (Tobler 1950: Plate CXLVI: 405-407)	
	Type 51 (Cat. 42/ FG 66)		Late Chalcolithic Palestinian origins (Strommenger et al. 2014:159)	/
b. Chaff-tempered fabtic but non-local shapes	Type 48 (Cat. 38/ FG 46)		/	/
	Type 55 (Cat.43/ FG 48-49)		/	/
C. Ambiguous Chaff-Tempered Pottery				
Description	Pottery Type (JA/HK)	Pottery Illustrations	Representative Local Chalcolithic Assemblage	Pottery Illustrations
Chaff-tempered fabric with indeterminate shapes	Type 57 (Cat. 43/ FG 62)		Amuq F/G (Braidwood and Braidwood 1960- Fig 180:18)	
	Type 67 (-/FG 47)		Unclear	Unclear
	Type 84 (-/ FG 64)		Amuq F/G (Braidwood and Braidwood 1960- Fig 180:13; Fig 231:8)	
	Type 71 (-/FG 50)		Unclear	Unclear
	Type 91 (-/FG 28 G 384)		Unclear	Unclear
	Type 79 (-/ FG 63)		Unclear	Unclear

Published data indicate that a partial statistical analysis of chaff-tempered pottery (corresponding to Groups A and C) has already been conducted at Habuba Kabira-Süd on a sherd basis. Building on this work, this present study extends the analysis to include Group B. In contrast, the excavation report for Jebel Aruda lacks comprehensive fabric descriptions and contains no systematic statistical analysis of its chaff-tempered pottery. This limitation precludes a precise frequency comparison of locally related pottery between the two sites. Nonetheless, a provisional assessment can be made on the basis of the available specimen descriptions.

According to the data, at Habuba Kabira-Süd, 6,113 sherds (Fabric Groups 55-67) are classified into chaff-tempered "private production" Group IV, accounting for 6.45% of the total assemblage (see Table 9 and Figure 17). At Jebel Aruda, the pottery descriptions identify chaff-tempered pottery in the following categories: 1 example of Type 28 (Cat. 15), 11 of Type 62 (Cat. 19), 9 of Type 31 (Cat. 20), 12 of Type 44 (Cat. 30), 1 of Type 48 (Cat. 38), and 6 of Cat. 43 (mainly Types 55/57). Taken together, these 40 examples represent 2.98% of the Jebel Aruda assemblage. Compared to Habuba Kabira-Süd, Jebel Aruda therefore contains a smaller proportion of chaff-tempered vessels.

However, this picture changes once hybrid forms belonging to Group B are taken into account. When included, the proportions of locally related pottery increase substantially at both sites. At Jebel Aruda, the total rises to 13.27%, comprising 40 chaff-tempered sherds, an additional 136 examples of Types 28/29, and 2 examples of Type 51, out of 1,341 vessels. At Habuba Kabira-Süd, the total reaches 7.44%, comprising 6,113 chaff-tempered sherds, 923 sherds of Type 28/29, 6 sherds of Type 75, 2 sherds of Type 77, and 1 examples of Type 51, from an assemblage of 94,754 sherds. On this revised basis, Jebel Aruda contains a larger proportion of locally related Chalcolithic pottery than Habuba Kabira-Süd.

Table 9: Comparative Frequency of Chaff-Tempered and Local Late Chalcolithic Vessels at Jebel Aruda and Habuba Kabira-Süd

Chaff Tempered Vessels/Sherds					
	Type (Group)	Quantity	Total Quantity	Total Assemblage (BRB and BT excluded)	Percentage
JA	28 (Cat. 15)	1	40	1,341	2.98%
	62 (Cat. 19)	11			
	31 (Cat. 20)	9			
	44 (Cat. 30)	12			

	48 (Cat. 38)	1			
	55/57 (Cat. 43)	6			
HK	Group IV	6,113	6,113	94,754	6.45%
Local Late Chalcolithic Related Vessels/Sherds					
	Type (Group)	Quantity	Total Quantity	Total Assemblage (BRB and BT excluded)	Percentage
JA	Chaff Tempered	40	178	1,341	13.27%
	Type 28/29 (Cat. 15/17)	136			
	Type 51 (Cat.42)	2			
HK	Group IV	6,113	7,045	94,754	7.44%
	Type 28/29 (FG 22.3/FG 22.4)	923			
	Type 75	6			
	Type 77	2			
	Type 51	1			

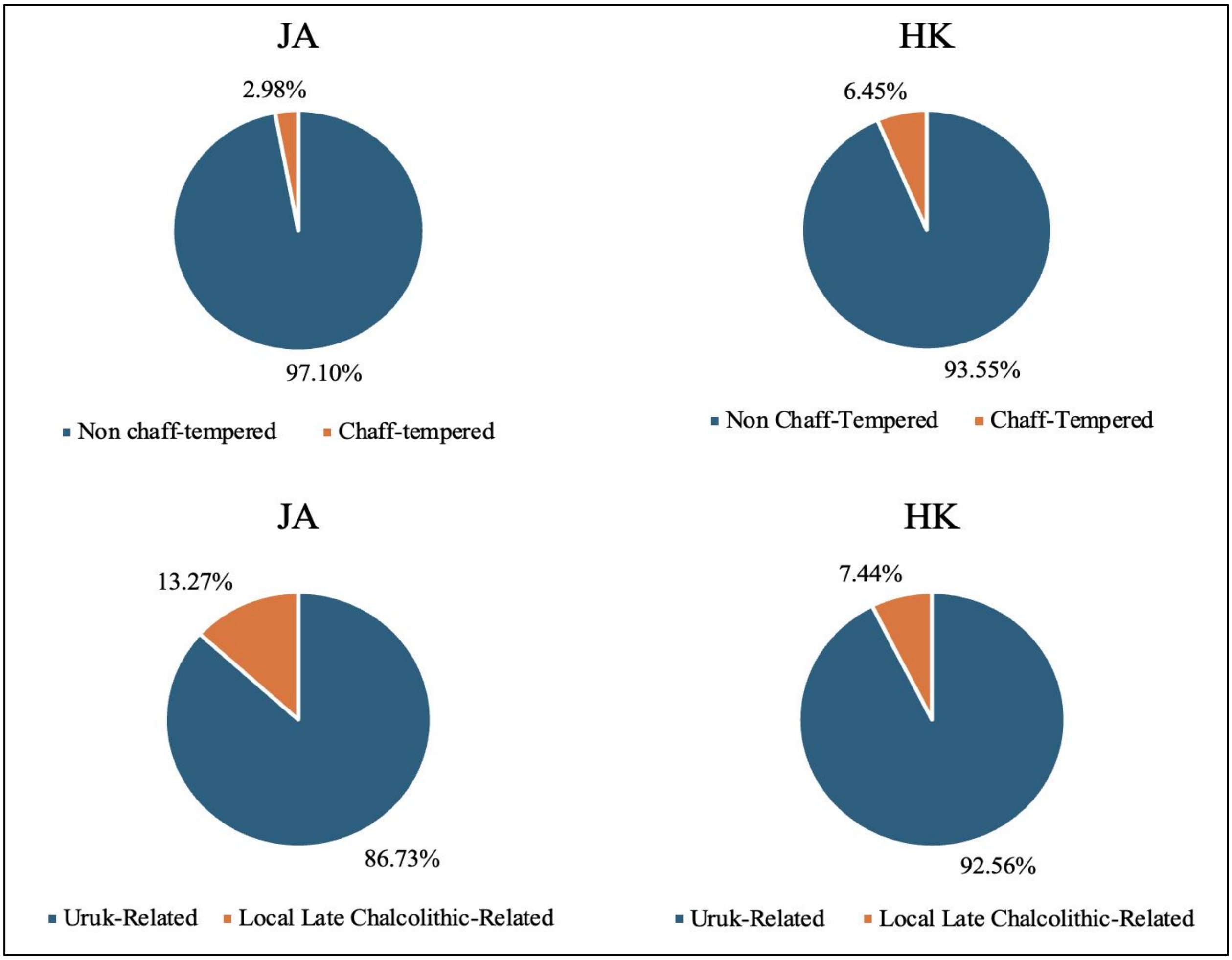


Figure 17: Percentage of Chaff-Tempered (upper) and Local Late Chalcolithic-Related Vessels (lower) at Jebel Aruda (JA) and Habuba Kabira-Süd (HK)

In summary, the analysis reveals a complex relationship between the sites and local Chalcolithic traditions. When focusing solely on chaff-tempered fabric, Habuba Kabira-Süd exhibits a higher percentage than Jebel Aruda. This, combined with the prevalence of unique forms at Habuba Kabira-Süd linked to local Chalcolithic traditions, emphasizes Habuba Kabira-Süd's distinctive status and its stronger connection to local traditions when compared with Jebel Aruda. However, when hybrid forms are considered, Jebel Aruda shows a higher proportion, primarily due to the significant presence of Type 28/29 vessels. This suggests that Jebel Aruda may have a stronger tie to local traditions if Type 28/29 represents an adoption or imitation of late Chalcolithic forms. Further investigation into the nature of Type 28/29 and its relationship with the local community is needed, though this question exceeds the scope of this study. Finally, it must be emphasized that these conclusions are preliminary, as they are constrained by the use of divergent registration systems at the two sites and the incomplete publication of fabric descriptions from Jebel Aruda. A reliable comparison hinges on the availability of comprehensive datasets employing a consistent methodological framework.

5. Concentration Comparison

To investigate pottery distribution patterns between the two sites, an exploratory data analysis was conducted. The analysis examined the spatial concentration and co-occurrence of vessel types, drawing on find locations recorded in excavation inventory lists. For accuracy, several categories of material were futher excluded: items lacking clear room-level provenance, and material from secondary contexts such as pits.

The inventory lists reveal significant differences in data quality between the two sites. At Jebal Aruda, 1,114 of 1,341 vessels and sherds (83.07%) have clearly documented find locations tied to specific rooms or contexts, making the site highly suitable for detailed distributional analysis. In contrast, at Habuba Kabira-Süd, only 479 of 1,292 vessels (37.07%) possess reliable location data. As a result, the distributional patterns observed at Habuba Kabira-Süd are less conclusive and require cautious interpretation. Nevertheless, the available evidence still provides valuable insights, which are considered in the following sections.

To analyze artifact distribution, distribution matrices were generated for each site using Python (see Tables 10 and 11). These matrices quantify the occurrence and co-occurrence of object types across distinct rooms or locations. Diagonal entries in the matrices represent the number of rooms containing a specific object type, while off-diagonal entries indicate the number of rooms where pairs of object types co-occur. For instance, at Jebel Aruda, Cat. 3, Cat. 4, and Cat. 5 pottery were documented in 53, 41, and 34 rooms, respectively, with Cat. 3 and Cat. 4 co-occurring in 21 rooms and Cat. 4 and Cat. 5 in 15 rooms. To enhance interpretability, matrix values are color-coded, with details provided in the Appendix A.

Concentration and co-occurrence analyses were conducted based on these matrices. The following section presents and discusses the results of the concentration analysis. Regarding co-occurrence, the relative completeness of the Jebel Aruda dataset enabled an exploratory study;

however, the insufficiency of the Habuba Kabira-Süd material precluded a comparable analysis. As the findings of the co-occurrence analysis are ancillary to the primary comparative framework of this dissertation, they have been detailed in Appendix C.

Table 10: Distribution Matrix of the Top 15 Most Widely Distributed Potteries at Jebel Aruda

	CAT 3	CAT 4	CAT 5	CAT 43	CAT 10	CAT 15	CAT 27	CAT 9	CAT 34	CAT 26	CAT 20	CAT 33	CAT 21	CAT 25	CAT 17
CAT 3	53	21	21	10	10	11	11	11	9	13	13	9	10	11	7
CAT 4	21	41	15	12	15	15	15	16	15	13	11	10	11	12	8
CAT 5	21	15	34	11	9	9	13	11	10	12	12	7	11	11	8
CAT 43	10	12	11	32	6	7	10	12	9	11	6	6	9	9	9
CAT 10	10	15	9	6	28	9	9	8	9	8	7	6	6	7	5
CAT 15	11	15	9	7	9	28	12	9	8	8	7	7	2	9	6
CAT 27	11	15	13	10	9	12	27	9	14	12	6	7	7	9	7
CAT 9	11	16	11	12	8	9	9	27	6	10	7	8	4	11	3
CAT 34	9	15	10	9	9	8	14	6	26	10	3	10	11	9	7
CAT 26	13	13	12	11	8	8	12	10	10	24	5	5	4	11	3
CAT 20	13	11	12	6	7	7	6	7	3	5	22	8	6	7	7
CAT 33	9	10	7	6	6	7	7	8	10	5	8	22	6	7	4
CAT 21	10	11	11	9	6	2	7	4	11	4	6	6	21	5	7
CAT 25	11	12	11	9	7	9	9	11	9	11	7	7	5	21	4
CAT 17	7	8	8	9	5	6	7	3	7	3	7	4	7	4	20

Table 11: Distribution Matrix of the Top 15 Most Widely Distributed Potteries at Habuba Kabira-Süd

	FG 27	FG 31	FG 41	FG 40	FG 8	FG 1	FG 36	FG 13	FG 22.1	FG 26	FG 24	FG 39.1	FG 3.2	FG 22.5	FG 23
FG 27	22	10	6	4	1	1	2	3	5	5	2	3	0	5	4
FG 31	10	21	11	4	2	3	3	3	5	3	2	3	1	6	1
FG 41	6	11	21	5	5	4	2	5	3	4	3	3	2	3	2
FG 40	4	4	5	15	2	1	2	1	1	1	1	1	1	3	2
FG 8	1	2	5	2	13	2	2	2	0	2	1	2	2	2	1
FG 1	1	3	4	1	2	12	2	2	1	2	1	3	0	0	0
FG 36	2	3	2	2	2	2	12	1	1	3	1	2	2	2	1
FG 13	3	3	5	1	2	2	1	11	1	3	2	1	2	2	0
FG 22.1	5	5	3	1	0	1	1	1	11	1	1	2	0	3	2
FG 26	5	3	4	1	2	2	3	3	1	10	2	2	1	1	2
FG 24	2	2	3	1	1	1	1	2	1	2	10	0	1	1	2
FG 39.1	3	3	3	1	2	3	2	1	2	2	0	10	0	2	1
FG 3.2	0	1	2	1	2	0	2	2	0	1	1	0	10	1	2

FG 22.5	5	6	3	3	2	0	2	2	3	1	1	2	1	9	1
FG 23	4	1	2	2	1	0	1	0	2	2	2	1	2	1	9

5.1 Concentration Metrics and the Comparative Analysis

The concentration analysis employs two metrics to evaluate the distribution of vessel types across rooms. The Distribution Breadth Index (DBI) measures the spatial spread of a type by recording the number of rooms in which it occurs, represented by the diagonal values of the matrix. The Dispersion Intensity Index (DII), by contrast, assesses the relationship between frequency and distribution by calculating the ratio of room count to total quantity for each type. Summary results for both indices are presented in Table 12 and Table 13, which lists the ten most and least concentrated categories, with the full dataset provided in Appendix B.

Table 12: Pottery Category Metrics and Associated Statistics for Jebel Aruda

Jebel Aruda Pottery Types with the 10 Highest DBI Values				
	Category (Type)	Distribution Breadth Index (DBI=Room Number)	Quantity	Dispersion Intensity Index (DII=Room Number/Quantity)
1	CAT 3 (Type 3)	53	141	0.38
2	CAT 4 (Type 4)	41	77	0.53
3	CAT 5 (Type 5)	34	57	0.60
4	CAT 43 (Include Type 52/59)	32	47	0.68
5	CAT 10 (Type 20-22)	28	36	0.78
6	CAT 15 (Type 28)	28	55	0.51
7	CAT 27 (Type 42)	27	51	0.53
8	CAT 9 (Type 10-13)	27	39	0.69
9	CAT 34 (Type 46)	26	49	0.53
10	CAT 26 (Type 38-41)	24	33	0.73
Jebel Aruda Pottery Types with the 10 Lowest DBI Values				
	Category (Type)	Distribution Breadth Index (DBI=Room Number)	Quantity	Dispersion Intensity Index (DII=Room Number/Quantity)
1	CAT 19 (Type 62)	1	11	0.09
2	CAT 38 (Type 48)	2	2	1.00
3	CAT 43 Hedgehog (Type 66)	3	3	1.00
4	CAT 22 (Type 33)	3	4	0.75
5	CAT 40 (Type 17)	5	5	1.00
6	CAT 35 (Type 47)	6	8	0.75
7	CAT 6 (Type 6)	7	8	0.88

8	CAT 13 (Type 64)	8	9	0.89
9	CAT 37 (Type 15)	8	10	0.80
10	CAT 12 (Type 24)	9	9	1.00

Statistical Data of Metrics			
	DBI	DII	Quantity
Mean	16.98	0.73	27.65
Standard Error	1.76	0.03	4.17
Median	14	0.75	20
Mode	14	1	15
Minimum	1	0.09	2
Maximum	53	1	141

Table 13: Pottery Category Metrics and Associated Statistics for Habuba Kabira-Süd

Habuba Kabira-Süd Pottery Types with the 10 Highest DBI Values				
	Group (Type)	Distribution Breadth Index (DBI=Room Number)	Quantity	Dispersion Intensity Index (DII=Room Number/Quantity)
1	FG 27 (Type 22)	22	30	0.73
2	FG 31 (Type 26)	21	40	0.53
3	FG 41 (Type 35)	21	31	0.68
4	FG 40 (Type 24)	15	15	1.00
5	FG 8 (Type 4)	13	14	0.93
6	FG 1 (Type 3)	12	14	0.86
7	FG 36 (Type 36)	12	19	0.63
8	FG 13 (Type 17)	11	12	0.92
9	FG 22.1 (Type 49)	11	12	0.92
10	FG 26 (Type 42/50)	10	11	0.91
Habuba Kabira-Süd Pottery Types with the 10 Lowest DBI Values				
	Group (Type)	Distribution Breadth Index (DBI=Room Number)	Quantity	Dispersion Intensity Index (DII=Room Number/Quanity)
1	FG 11.1 (Type 70)	1	1	1.00
2	FG 11.3 (Type 75)	1	1	1.00
3	FG 3.1 (Type 72)	1	1	1.00
4	FG 3.4 (Type 10)	1	1	1.00
5	FG 34 (Type 43)	1	1	1.00
6	FG 37.1(Type 24)	1	1	1.00
7	FG 43(Type 33)	1	1	1.00
8	FG 56 (Type 56)	1	1	1.00
9	FG 61 (Type 23)	1	1	1.00
10	FG 64 (Type 85)	1	1	1.00

Statistical Data of Metrics			
	DBI	DII	Quantity

Mean	5.72	0.92	6.88
Standard Error	0.60	0.01	0.89
Median	4	1	4
Mode	1	1	1
Minimum	1	0.525	1
Maximum	22	1	40

Statistical analysis on DBI reveals that a pottery type at the Jebel Aruda is found in an average of 16.98 rooms (see Table 12). Categories associated with eating and drinking functions, such as Cat. 3 (Type 3 small goblets), Cat. 4 (Type 4 beaked rim bowls), and Cat. 5 (Type 5 small round bowls), exhibit the broadest distribution, occurring in the greatest number of rooms. In contrast, rare forms, such as Cat. 19 (Type 62 chaff tempered cooking pots), Cat. 38 (Type 48 oval vessels with a narrow spout on the top), Cat. 22 (Type 33 jars with straight spout), and Cat. 43 hedgehog (Type 66) , are present in the fewest. These results are consistent with the frequency analysis. Both show that abundant pottery forms appear in multiple rooms, while rare forms have a limited distribution. At Habuba Kabira-Süd, the DBI reveals that pottery is typically found in an average of 5.72 rooms (see Table 13). Specifically, FG 27 (Type 22 vessels with lugs), FG 31 (Type 26 Uruk red), and FG 41 (Type 35 flat-bottomed vases with cylindrical necks) are found in more than 20 rooms each, far exceeding the remainder, with the next highest category appearing in 15 rooms. Conversely, rare forms, including FG 11.1 (Type 70), FG 11.3 (Type 75), and FG 3.1 (Type 72), are restricted to a single room. As at Jebel Aruda, the distribution broadly reflects frequency ranking.

Of particular interest are the DII results, which reveal significant variation in pottery concentration. At Jebel Aruda, values range from highly concentrated forms (as low as 0.09) to highly dispersed ones (up to 1.0). A scatter plot of DII against DBI (Figure 18) demonstrates their broadly inverse relationship. This visualization highlights that, by incorporating quantity, the DII provides a more complete measure of concentration compared to the DBI. Many vessel types, though distributed across numerous rooms, are in fact highly localized once frequency is taken into account. The plot further shows that most categories cluster at the higher end of the DII range, indicating relatively uniform spatial distributions. One form, however, disrupts this pattern: Cat. 19 (Type 62). With a DII of 0.09, it is statistically identifiable as an outlier, reflecting unusual concentration within a restricted set of rooms or activity areas. Two additional categories also register distinctly low values: Cat. 33 (Type 45, DII=0.29) and Cat. 3 (Type 3, DII=0.38). Quantity comparisons (Figure 19) clarify their distinct behaviors: Cat. 19 is rare but highly localized; Cat. 33 is moderately abundant but clustered; and Cat. 3, while numerous and widespread, is anomalously concentrated in specific contexts. On this chart, Cat. 3, alongside Cat. 19, also separates as an outlier. Conversely, forms with a DII of 1.0, indicating maximal dispersion, are predominantly rare (quantities ≤10). Among these, Cat. 29 is particularly significant: all ten specimens were recovered from ten different rooms. This wide distribution may indicate a specialized item used on particular occasions and intentionally scattered across multiple activity areas. Relative to Jebel Aruda, the Habuba Kabira-Süd data exhibits a more scattered pattern, with multiple groups widely dispersed. Nevertheless, a distinct cluster comprising FG 31 (Type 26) , FG 41 (Type 35) , and FG 27 (Type 22) occupies the far right end of both Figure 18 and Figure 19, indicating a unique distributional signature. FG 31, in particular, with high quantity and low DII, closely parallels Cat. 3 at Jebel Aruda.

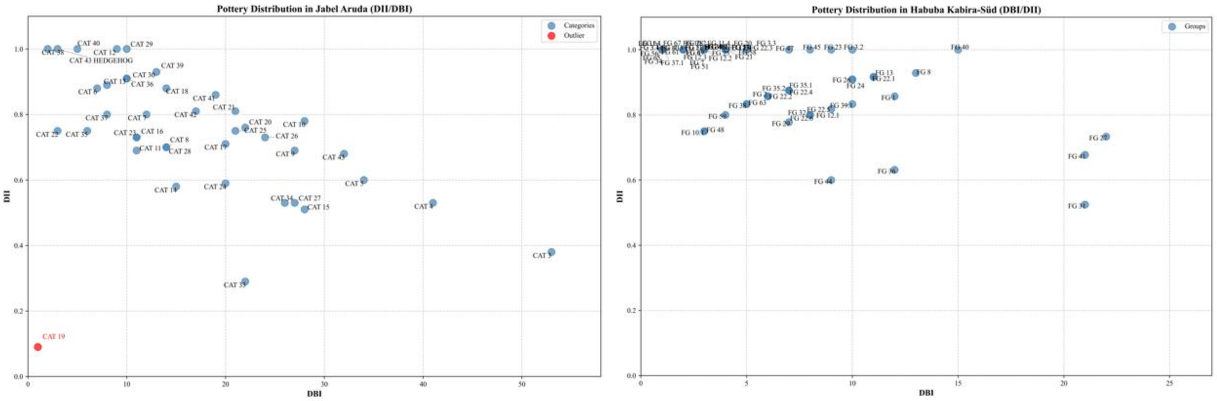


Figure 18: Scatter Plot of DII against DBI for Jebel Aruda (left) and Habuba Kabira-Süd (right)

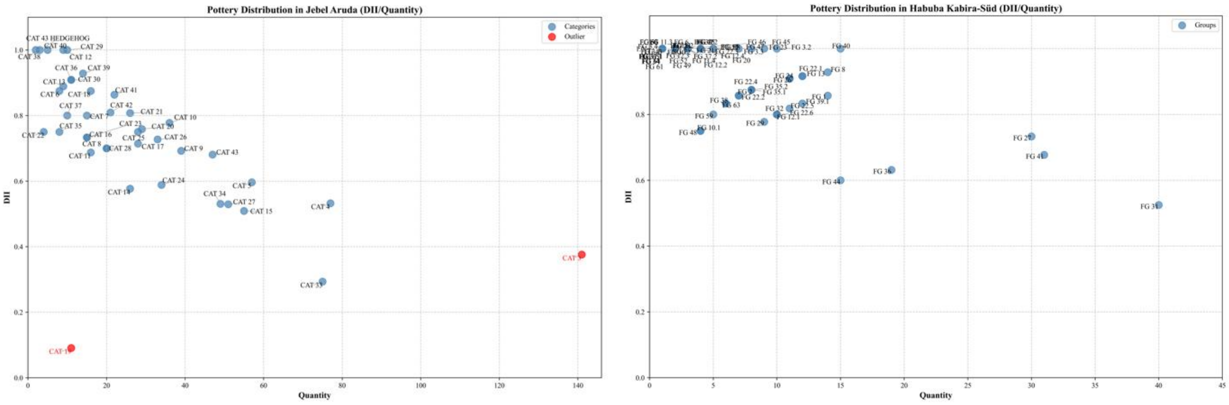


Figure 19: Scatter Plot of DII against Quantity for Jebel Aruda (left) and Habuba Kabira-Süd (right)

5.2 Distribution Types and Distribution Comparison Between the Sites

Building on the distributional patterns outlined above, a further grouping of vessel types was undertaken for both sites using DII values and quantity. This analysis excluded pottery forms represented by only a single specimen, as quantitative measures of concentration and spatial distribution for such types are not statistically meaningful.

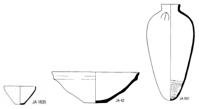
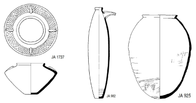
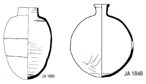
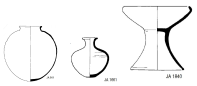
Based on these two variables (DII and quantity), the vessel types at both sites could be categorized into four distributional groups, reflecting their abundance and spatial patterning (see Table 14, Table 15 and Appendix B for the full classification):

1. Abundant and concentrated — high quantity with low DII
2. Abundant and dispersed — high quantity with moderate to high DII.
3. Rare and concentrated — low quantity with low to moderate DII
4. Rare and dispersed — low quantity with high DII.

This classification provides a framework for comparing the assemblages from Jebel Aruda and Habuba Kabira-Süd. However, as noted earlier, limitations in the data quality for Habuba Kabira-Süd render its results less conclusive than those for Jebel Aruda. To mitigate this issue for Habuba Kabira-Süd, a supplementary item-by-item spatial mapping was carried out for selected types with distinctive distributional patterns, as well as for several rare forms. The results of these comparative analyses are presented below.

Table 14: Classification of the Jebel Aruda Pottery Assemblage by Abundance and Spatial Distribution

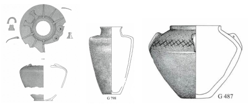
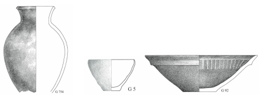
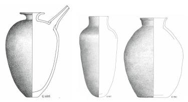
	Distribution Group	Criteria [(1)]	Group Composition and Metrics for Key Categories (DBI, Quantity, DII)	Illustrations of Key Categories

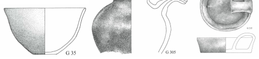
1	Abundant and Concentrated	High quantity (>25.10), low DII (0.4-0.71)	Cat. 3*/33/4/5/15/27/34/43/9/24/14 - Cat. 3* (53 rooms, 141 items, DII =0.38) - Cat. 33 (22 rooms, 75 items, DII =0.29) - Cat. 4 (41 rooms, 77 items, DII =0.53)	
2	Abundant and Dispersed	High quantity (>25.10), high/moderate DII (>0.71)	Cat. 10/26/20/25/21/17 - Cat. 10 (28 rooms, 36 items, DII= 0.78) - Cat. 26 (24 rooms, 34 items, DII= 0.76) - Cat. 21 (21 rooms, 26 items, DII= 0.81)	
3	Rare and Concentrated	low quantity (<25.10), low/moderate DII (<0.86)	Cat. 19*/42/28/8/7/11/23/16/37/35/22 - Cat. 19* (1 rooms, 11 items, DII =0.09) - Cat. 41 (19 rooms, 22 items, RCI 0.86)	
4	Rare and Dispersed	Low quantity (<25.10), high DII (>0.86)	Cat. 29/ 41/18/39/36/30/29/12/13/6/ 40/Hedgehog /38 - Cat. 29 (10 rooms, 10 items, DII= 1.00) - Cat. 12 (9 rooms, 9 items, DII= 1.00) - Cat. 39 (13 rooms, 14 items, DII =0.93)	

(1) The criteria for high/low pottery quantity use the dataset's average (25.10), with DII thresholds for high (≥0.86, top 33.3%) and low (≤0.71, bottom 33.3%), excluding outlers.

(2) Asterisk (*) denotes statistical outliers

Table 15: Classification of the Habuba Kabira-Süd Pottery Assemblage by Abundance and Spatial Distribution

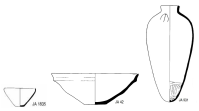
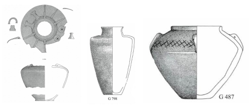
	Distribution Group	Criteria (1)	Group Composition and Metrics for Key Categories (DBI, Quantity, DII)	Illustrations of Key Categories
1	Abundant and Concentrated	High quantity (>6.70), low DII (≤0.80)	FG 27*/31*/41*/36/44/12.1/22.6/32/29 - FG 31* (21 rooms, 40 items, DII =0.53) - FG 41* (21 rooms, 31 items, DII =0.68) - FG 27* (22 rooms, 30 items, DII =0.73)	
2	Abundant and Disperse	High quantity (>6.70), high/moderate DII (>0.80)	FG 40/1/8/13/22.1/39.1/22.5/24/26/3.2/ 23/22.4/35.1/35.2/45/2/22.2/47 - FG 40 (15 rooms, 15 items, DII= 1) - FG 1 (12 rooms, 14 items, DII= 0.86) - FG 8 (13 rooms, 14 items, DII= 0.93)	
3	Rare and Concentrated	low quantity (<6.70), low/moderate DII (<0.875)	FG 38/63/59/10.1/48 - FG 38 (5 rooms, 6 items, DII =0.83) - FG 63 (5 rooms, 6 items, DII =0.83) - FG 59 (4 rooms, 5 items, DII =0.8)	

4	Rare and Disperse	Low quantity (<6.70), high DII (≥0.875)	FG 3.3/22.3/15/57/12.4/21/58/20/12.2/ 37.2/39.2/46/11.4/25/42/11.2/51/10.2/ 12.3/28/49/52/6/7 - FG 3.3 (5 rooms, 5 items, DII= 1.00) - FG 22.3 (5 rooms, 5 items, DII= 1.00) - FG 15 (4 rooms, 4 items, DII= 1.00)	

(1) The criteria for high/low pottery quantity use the dataset's average (6.70), with DII thresholds for high (≥0.875, top 33.3%) and low (≤0.80, bottom 33.3%), excluding outliers and potteries of singular occurrences.

(2) Asterisk (*) denotes statistical outliers.

5.2.1 Distribution Type 1: Abundant and Concentrated Forms

	Group Composition and Metrics for Key Categories (DBI, Quantity, DII)	Illustrations of Key Categories
JA	Cat. 3*/33/4/5/15/27/34/43/9/24/14 - Cat. 3* (53 rooms, 141 items, DII =0.38) - Cat. 33 (22 rooms, 75 items, DII =0.29) - Cat. 4 (41 rooms, 77 items, DII =0.53)	
	Group Composition and Metrics for Key Categories (DBI, Quantity, DII)	Illustrations of Key Categories
HK	FG 27*/31*/41*/36/44/12.1/22.6/32/29 - FG 31* (21 rooms, 40 items, DII =0.53) - FG 41* (21 rooms, 31 items, DII =0.68) - FG 27* (22 rooms, 30 items, DII =0.73)	

A comparison of Distribution Type 1 (Abundant and Concentrated) reveals a stark contrast in group composition between Jebel Aruda and Habuba Kabira-Süd. At Jebel Aruda, this group encompasses a wide functional range, including eating and drinking vessels (Cat. 3, 4, 5, 9), food preparation pots (Cat. 15), storage vessels (Cat. 24, 27, 33, 34), and luxury Uruk red ware (Cat. 14). By contrast, Group 1 at Habuba Kabira-Süd comprises nine vessel types, three of which (FG 27, FG 31, and FG 36) correspond to forms also classified within Group 1 at Jebel Aruda (Cat. 10, Cat. 14, and Cat. 24/33). The remaining six types, however, fall into Distribution Type 2-4 at Jebel Aruda, revealing divergence in grouping patterns between the two sites. Moreover, four types (FG 27, FG 31, FG 29, and FG 44) out of nine represent potential luxury pottery, suggesting an emphasis on socially valued or specialized forms.

Within these broader compositions, distinct forms emerge at each site. At Jebel Aruda, the outlier Cat. 3 goblets (Type 3) are particularly notable: they are numerically dominant and display a unique distributional pattern, a prominence not paralleled at Habuba Kabira-Süd. Of the 141 specimens with secure provenience, only four derive from the northern houses (one each from NA, NB, NF, and NH), and a single example comes from the temple (see Figure 20). The remaining 137 specimens originate from the southern area. Within this area, Building SI contains the greatest concentration, accounting for 52% (n = 74) of the total. Except for the central court and a few side rooms, Cat. 3 goblets are present in nearly every room of the building, with Room S7 representing the locus of the highest density (n=17). By contrast, the distribution of Type 3 goblets at Habuba Kabira-Süd is far less concentrated and is categorized under Distribution Type 2. Pottery with secure provenience is scattered across seven northern rooms, two southern rooms, and the Qannas Gate. Even when vessels with uncertain provenience are included, their maximum possible presence spans 61 rooms across 19 buildings and the 4 rooms at the Qannas Gate, with no room containing more than five examples (See Figure 21). This contrast, combined with frequency analysis demonstrating that goblets are numerically dominant only at Jebel Aruda, suggests a site-specific demand concentrated in the southern sector of Jebel Aruda. The evidence invites several interpretive possibilities. Is this linked to the culinary practices of a distinct group within the southern residences who favored the use of small cups? Could these vessels have served as social markers for high-status individuals, consistent with the special status of Building SI? Alternatively, might this pattern relates to the proposed religious character of the site itself?

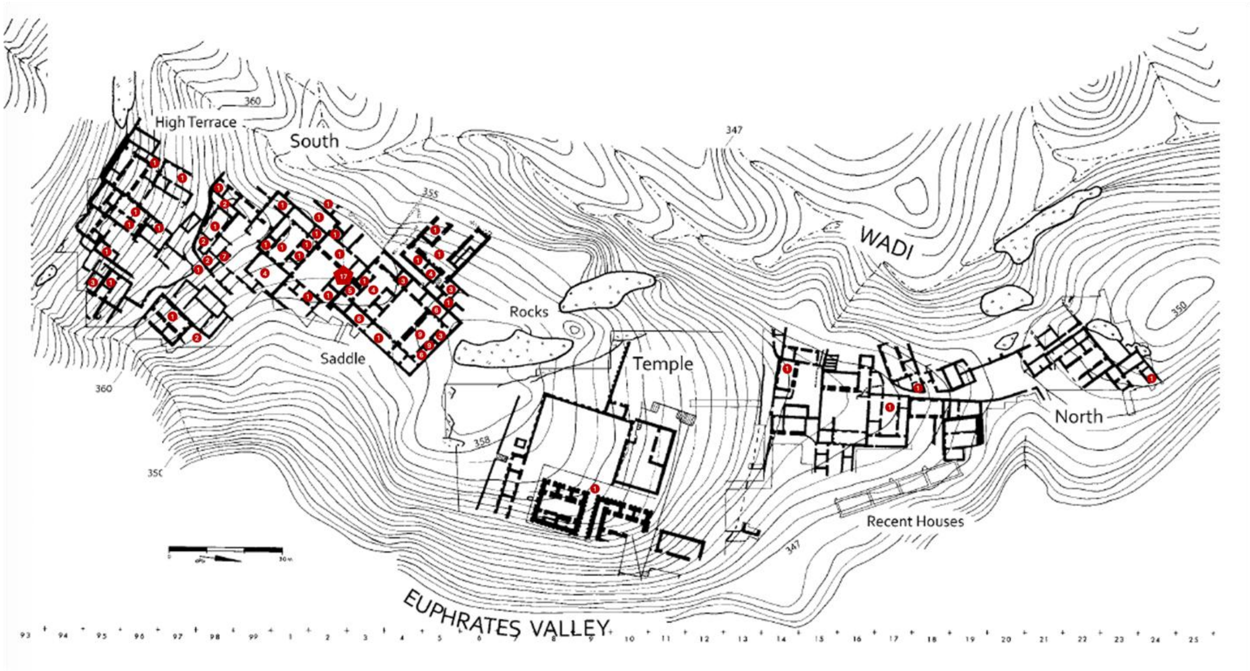

High Terrace
South
WADI
Rocks
Saddle
Temple
North
Recent Houses
EUPHRATES VALLEY
360
355
347
350
358

Figure 20: Spatial Distribution of Type 3 Goblets at Jebel Aruda

Figure 21: Spatial Distribution of Type 3 Goblets at Habuba Kabira-Süd - Definite Findspots with Quantities (Left) and Possible Presence (Right)

At Habuba Kabira-Süd, by contrast, FG 31 (Type 26 Uruk red vessels) represent the most distinct element of Group 1. This form occurs in high quantity and exhibits a low DII value (0.53), indicating a concentrated distribution. With a quantity of 60 specimens, the type ranks second in terms of quantity according to frequency analysis. Among 60 specimens, 40 have a secure location. Of these, 10 derive from Room e of Building Anw 17, accounting for 25% of the total (see Figure 23). The remaining 29 are distributed across 20 rooms in 10 buildings, with one additional example recovered from a tower chamber in the city wall. When vessels of uncertain provenience are included, the pattern suggests a markedly clustered distribution, concentrated particularly around Building Anw 17 in the north and Buildings Anw 31 and Anw 35 in the south. At Jebel Aruda, Uruk red ware (Cat. 14) is also included in Group 1, but it is neither especially abundant nor distinctive (see Figure 22). From a total of 26 specimens, four derive from Room S51 in Building SIII and another four from Room S90 in Building SV, while the remainder are scattered across three northern rooms (n = 4) and sixteen southern rooms (n = 14). The distinctive clustering and relative abundance of FG 31 at Habuba Kabira-Süd may point to unprecedented patterns of use at the site. If Uruk red ware functioned as a marker of social status, its concentration around particular buildings could reflect more pronounced social stratification within Habuba Kabira-Süd than is evident at Jebel Aruda. In addition, its abundance may signal access to distinctive sources of wealth that differentiated Habuba Kabira-Süd's community from that of Jebel Aruda.

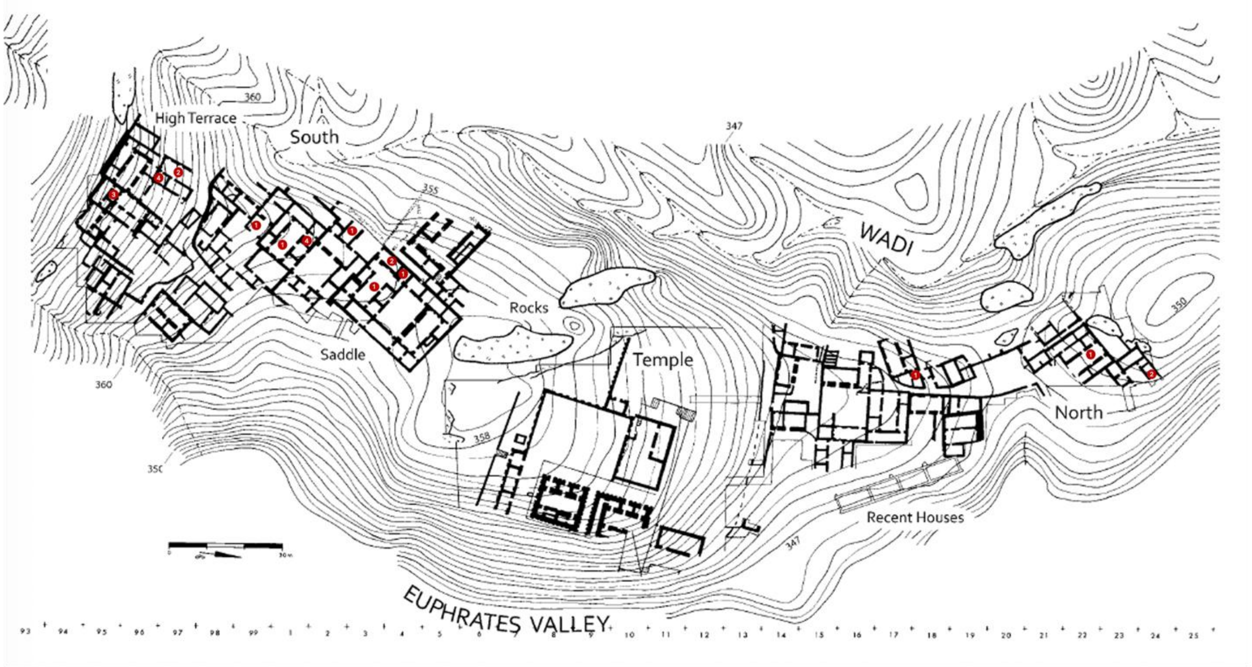
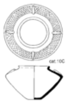
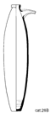
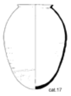
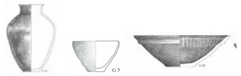


Figure 22: Spatial Distribution of Type 26 Uruk Red Vessels at Jebel Aruda

Figure 23: Spatial Distribution of Type 26 Uruk Red Vessels at Habuba Kabira-Süd - Definite Findspots with Quantities (Left) and Possible Presence (Right)

5.2.2 Distribution Type 2: Abundant and Dispersed Forms

	Group Composition and Metrics for Key Categories (DBI, Quantity, DII)	Illustrations of Key Categories
JA	Cat. 10/26/20/25/21/17 - Cat. 10 (28 rooms, 36 items, DII= 0.78) - Cat. 26 (24 rooms, 34 items, DII= 0.76) - Cat. 21 (21 rooms, 26 items, DII= 0.81)	  
	Group Composition and Metrics for Key Categories (DBI, Quantity, DII)	Illustrations of Key Categories

HK	FG 40/1/8/13/22.1/39.1/22.5/24/26/3.2/23/22.4/35.1/35.2/45/2/22.2/47 - FG 40 (15 rooms, 15 items, DII= 1) - FG 1 (12 rooms, 14 items, DII= 0.86) - FG 8 (13 rooms, 14 items, DII= 0.93)	

Group 2 comprises abundant and widely dispersed vessels. At Jebel Aruda, this group includes six categories: Cat. 10 (Types 20-22 small vessels with shoulder lugs and decoration), Cat. 26 (Types 38- 41 elongated spouted vessels), Cat. 20 (Type 31 baggy jars), Cat. 25 (Type 37 drooping spouted vessels), Cat. 21 (Type 32 bottle-shaped vessels), and Cat. 17 (Type 29 egg-shaped pots). With the exception of Cat. 10, these categories largely consist of storage vessels. The statistical results align with domestic contexts, suggesting that these forms fulfilled everyday needs and were widely employed across the site.

At Habuba Kabira-Süd, Group 2 is more extensive, encompassing 18 forms but sharing only a single category with Jebel Aruda: FG 39.1 (Types 38-40, corresponding to part of Cat. 26). The forms is the characteristic Late Uruk narrow elongated vessel with spouts. At Jebel Aruda, this form is dispersed across 24 rooms from 33 specimens, while at Habuba Kabira-Süd it occurs in 10 rooms from 12 specimens. The comparable distribution at both sites suggests that this vessel type was in high demand and regularly used in daily activities across multiple contexts. The remaining Group 2 forms at Habuba Kabira-Süd differ significantly from the Jebel Aruda assemblage. Excluding FG 40 (a broad category of flat-bottomed vases with funnel-shaped necks), the most prominent examples are FG 1 (Type 3 small goblet cups) and FG 8 (Type 4 beaked-rim bowls). These two forms rank among the most abundant at the site, exhibiting DIIs of 0.86 and 0.93 respectively. Their broadly dispersed distributions contrast sharply with patterns at Jebel Aruda, where such forms are more concentrated. Similar to the pattern observed with FG 31, this disparity further highlights the divergent ceramic assemblages of Habuba Kabira-Süd and Jebel Aruda.

In addition, Group 2 at Habuba Kabira-Süd includes two forms absent at Jebel Aruda: FG 47 (Type 67 pottery stands) and FG 45 (Type 82 elongated bottles). The limited quantity of these finds allows for detailed analysis, revealing a significant distribution pattern for the FG 47 stands. These stands were found in ten buildings, with a clear concentration in nine structures in the northeastern sector, the earliest part of the settlement by chronology (Table 16 and Figure 24). In contrast, the southwestern sector contained only a single example, located in the prestigious building Anw 40.

Table 16: Distribution of Pottery Stands across Buildings and Rooms

Area	Building	Room/Location
Northeast	Anw 1	Open space f
	Anw 2	Room i/j
	Anw 3	Room g/o/p; Open space j
	Anw 4	Room a
	Anw 6	Room b/e/f; Open space d
	Anw 7	Room i
	Anw 8	Room a/c/d; Open space b
	Anw 9	Room e
	Anw 13	Anw 13 R g
Southwest	Anw 40	Room b/g/h

Figure 24: Spatial Distribution of Type 67 Pottery Stand at Habuba Kabira-Süd - Definite Findspots with Quantities (Left) and Possible Presence (Right)

Anw 40, situated in Plot 23 along the main "long street," formed part of a substantial residential complex possibly linked with Anw 38. Vallet (1997: Page 55) suggests it likely housed higher-status families who expanded their living space by encroaching on areas previously belonging to plots 19 and 22 during Phase 2. This tripartite building features an eastward-facing entrance, a niched central hall, and seven side rooms (see Figure 25). Only a few structures, such as Anw 41 and the temple on Tall Qannas, exhibit comparable or richer decoration. Sherds of FG 47 were recovered in Rooms b and h, and a complete vessel (G886) was discovered in the secluded Room

g. This room also yielded a large flint blade, a copper/bronze spearhead and pin, and and pottery vessels G 556, G741, G 794, G 809 and G 947 (see Figure 26). On the basis of vessel morphology, it is plausible that the stand was intended to support G 947, the only vessel in the assemblage with a round base requiring stabilization, or alternatively, perishable objects that have not survived.

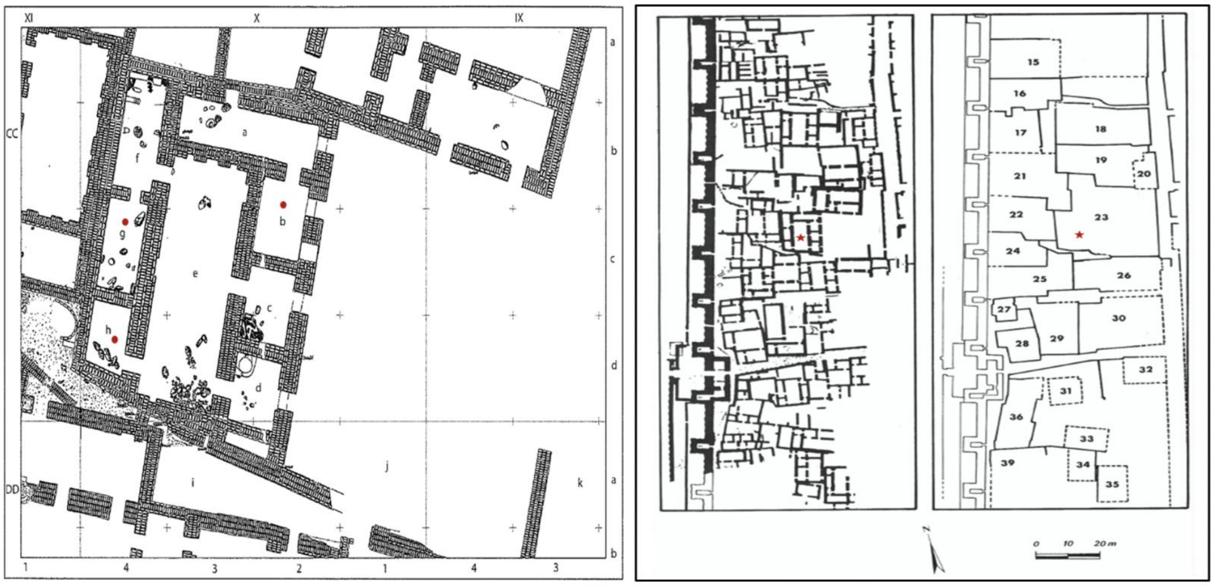


Figure 25: Architectural Plan (left) and Location (right) of Anw 40 (Edited Based on Kohlmeyer and Ludwig 2021: Figure 38 and Vallet 1997: Figure 5)

The distinct status of Anw 40, the seclusion of Room g, and the presence of rare artifacts suggest that pottery stands likely served specialized or symbolic functions. Statistical analysis in Chapter 3 confirms their presence at Habuba Kabira-Süd is not a random occurrence. Their exclusive presence at Habuba Kabira-Süd indicates inter-site differences. This may reflect Habuba Kabira-Süd's role as a large settlement and potential trade center, accommodating diverse ceramic forms and functions or a distinct cultural group favoring pottery stands. Conversely, Jebel Aruda's more focused activities may not have required such forms. These interpretations, however, remain tentative pending further evidence.

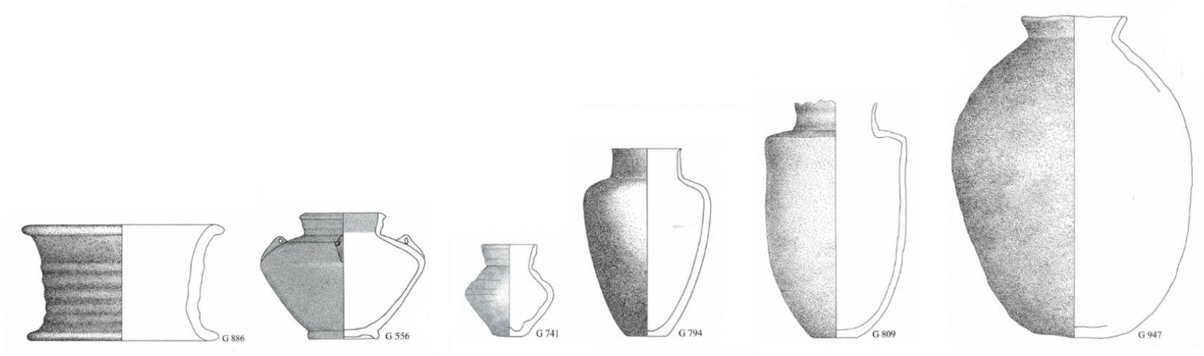
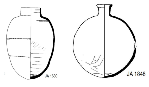
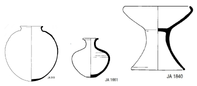
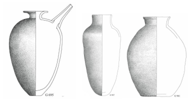
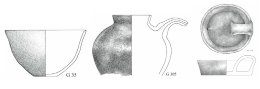


Figure 26: Pottery Stand G886 and Associated Vessels from Room G of Anw 40

5.2.3 Distribution Type 3 and 4: Rare and Concentrated / Rare and Dispersed Forms

	Distribution Type	Group Composition and Metrics for Key Categories (DBI, Quantity, DII)	Illustrations of Key Categories
JA	Rare and Concentrated	Cat. 19*/42/28/8/7/11/23/16/37/35/22 - Cat. 19* (1 rooms, 11 items, DII =0.09) - Cat. 41 (19 rooms, 22 items, RCI 0.86)	
	Rare and Dispersed	Cat. 29/ 41/18/39/36/30/29/12/13/6/40/Hedgehog /38 - Cat. 29 (10 rooms, 10 items, DII= 1.00) - Cat. 12 (9 rooms, 9 items, DII= 1.00) Cat. 39 (13 rooms, 14 items, DII =0.93)	
	Distribution Type	Group Composition and Metrics for Key Categories (DBI, Quantity, DII)	Illustrations of Key Categories
HK	Rare and Concentrated	FG 38/63/59/10.1/48 - FG 38 (5 rooms, 6 items, DII =0.83) - FG 63 (5 rooms, 6 items, DII =0.83) FG 59 (4 rooms, 5 items, DII =0.8)	
	Rare and Dispersed	FG 3.3/22.3/15/57/12.4/21/58/20/12.2/37.2/39.2/ 46/11.4/25/42/11.2/51/10.2/12.3/28/49/52/6/7 - FG 3.3 (5 rooms, 5 items, DII= 1.00) - FG 22.3 (5 rooms, 5 items, DII= 1.00) - FG 15 (4 rooms, 4 items, DII= 1.00)	

Pottery classified under Distribution Type 3 (Rare and Concentrated) and Distribution Type 4 (Rare and Dispersed) primarily consists of luxury items and other uncommon forms at Jebel

Aruda. At Habuba Kabira-Süd, however, the available dataset is more limited, and these groups also include ordinary pottery types whose numbers are reduced by imprecise room attribution. In both cases, the small sample size renders the DII highly sensitive to quantitative variation, producing abrupt fluctuations. Consequently, the distinction between "concentrated" and "dispersed" distributions is not always clear, making a combined analysis of these two groups necessary.

At Jebel Aruda, Distribution Type 3 includes several categories, such as Cat. 11 (Type 23 "perfume bottles"), Cat. 42 (Type 50 vessels with ears and handles), Cat. 35 (Type 47 side-rolled rim vessels), and Cat. 16 (Type 60 calcite-tempered cooking pots). The outlier, Cat. 19, also belongs to this group. It possesses the lowest DII among all categories, with all 11 items originating from a single room. Distribution Type 4 is characterized primarily by more intriguing forms such as Cat. 29 (Type 61 blister pots), Cat. 41 (Type 49 miniatures), Cat. 39 (Type 16 "fruit stands"), Cat. 40 (Type 17 dishes with small cups on the rim), and hedgehog-shaped vessels (Type 66).

At Habuba Kabira-Süd, Group 3 is the smallest of the four distribution groups, comprising only five types. These include FG 63 (Type 63 round-bottomed bottles) and FG 48 (Type 55 socket pipes) used in drainage systems, as well as three forms unique to Habuba Kabira-Süd: FG 10.1 (Type 68 round-bottomed steep-walled bowls), FG 38 (Type 80 flat-bottomed bottles with long straight spouts) , and FG 59 (Type 78 flat-bottomed pots with cylindrical necks and beaded rims). None of these overlap with the forms represented at Jebel Aruda. By contrast, Group 4 encompassing a wide range of vessels from small bowls (FG 11.2/Type 7) and pots (FG 22.3/Type 28) to large storage vessels (FG 28/Type 92), vases (FG 42/Type 83), and distinctive types such as bowls with ear handles on the rim (FG 12.3/Type 74). It also includes tools (FG 51-52/Type 71) that appear to have been specialized implements used across many rooms. This group shares four common forms with Jebel Aruda: ladles (FG 6/Type 6), fruit stands (FG 12.4/Type 16), basins (FG 15/Type 14), and oval vessels with spouts on the top (FG 46/Type 48).

The fluid nature of ceramic classification poses challenges for comparative analysis and further investigation. Nevertheless, Cat. 19 vessels at Jebel Aruda represent a notable exception that merits detailed examination. With an exceptionally low DII (0.09), they form a complementary, mirroring pair with pottery stands from Habuba Kabira-Süd: Cat. 19 vessels are highly concentrated at Jebel Aruda but absent at Habuba Kabira-Süd, while the pottery stands show the opposite pattern, being prevalent at Habuba Kabira-Süd and absent at Jebel Aruda. Moreover, the association of Cat. 19 vessels with the Amuq F culture enhances their archaeological significance. This unique distribution warrants a thorough investigation of their archaeological context.

This investigation reveals that all of these chaff-tempered "organic cooking vessels" were recovered exclusively from Court N18 within House NC (see Figure 27). As the principal complex in the site's northern sector, House NC is organized around Court N18 and is flanked by a western reception area, northern domestic quarters, and an eastern workroom or kitchen (Room N13). The presence of two large hearths and sixteen smaller cooking installations indicates that Court N18 primarily functioned as a food preparation area. One complete Cat. 19 vessel, accompanied by a mud lid, was found in one hearth, with fragments of another on the floor. If the Category 19 artifacts are indeed cooking vessels, their exclusive deposition in Court N18 raises two possible interpretations: (1) specialized food preparation in a dedicated space, or (2) a distinct culinary tradition tied to a specific social or ethnic group. This pattern also presents an instructive contrast with findings at Hacınebi, where it is Uruk Mesopotamian material that is spatially concentrated within a local Chalcolithic settlement (Stein, 2002). At Jebel Aruda, however, it is a locally-produced yet unique ceramic type that is segregated. This uniqueness prompts critical questions regarding the socio-functional character of Jebel Aruda, and particularly its northern sector. Could this concentration indicate that a specialized group resided in or operated from House NC, perhaps for religious, ritual, or other culturally-specific purposes at Jebel Aruda?

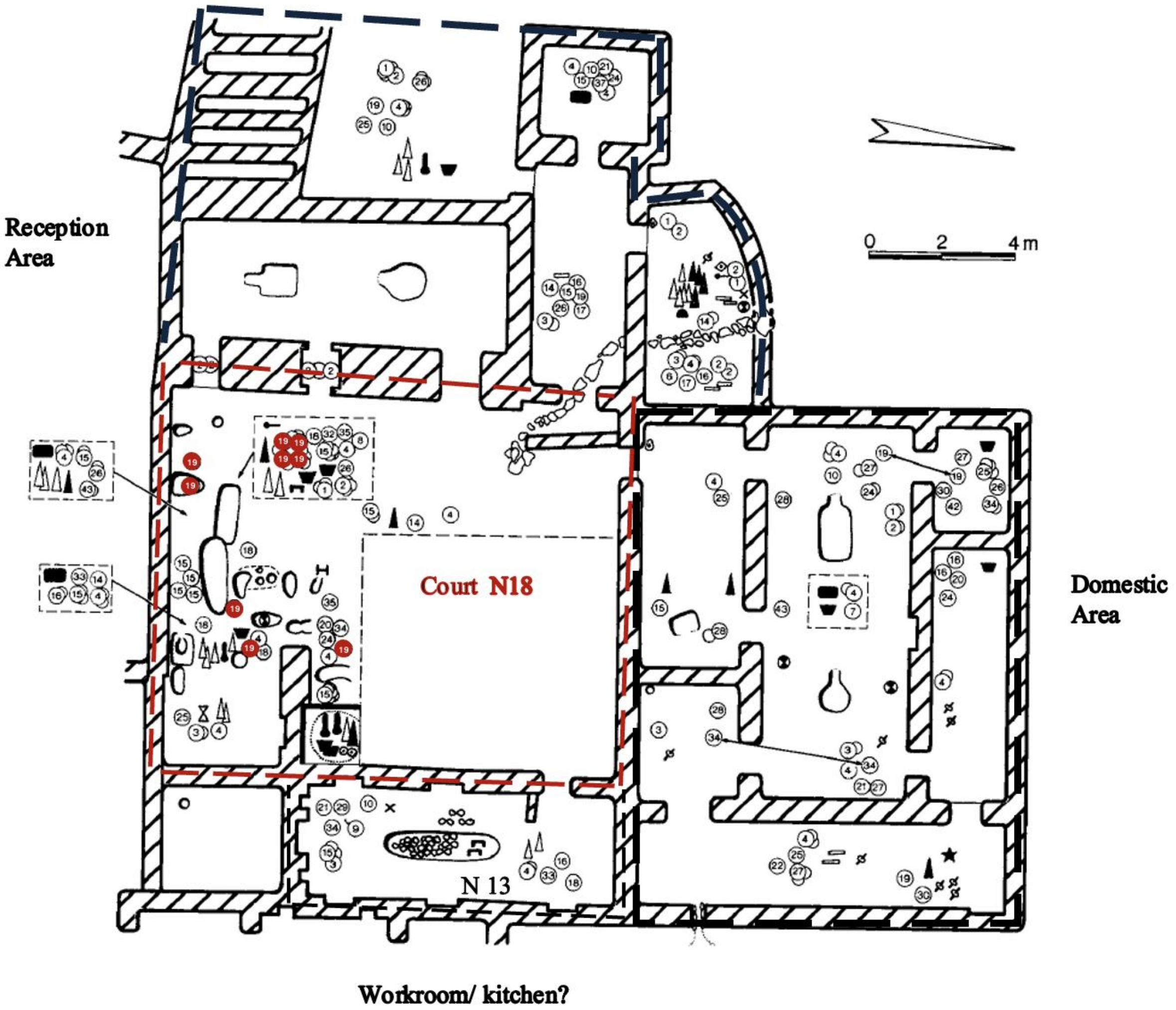


Figure 27: Cat. 19 and Other Room Contents in House NC (Edited based on Van Driel+ and Van Driel-Murray 2023: Figure 4.36)

In summary, the distribution metrics DBI and DII provide valuable insights into the ceramic assemblages at both sites. The DBI results parallel those of the frequency analysis: the most abundant forms tend to occur in a larger number of rooms, suggesting a relatively uniform distribution pattern. When quantitative factors are taken into account, the DII reveals further distinctions. Combined with DBI and quantity data, the DII identifies two statistical outliers at Jebel Aruda and three at Habuba Kabira-Süd. The distribution groups derived from DII demonstrate differences between the two sites. Group 1 shares three common forms but diverges in others; Group 2 shares one; Group 3 shares none; and Group 4 shares four forms but differs otherwise. Some of this variation may reflect data quality issues, which cannot be entirely ruled out. Nevertheless, one conclusion is clear: at Habuba Kabira-Süd the most concentrated forms are luxury vessels, whereas at Jebel Aruda they are predominantly eating and drinking pottery. The best example of this contrast is provided by Cat. 3 goblets and FG 26 Uruk Red ware, two of the most abundant and concentrated forms, one an eating and drinking vessel, the other a luxury vessel. In addition, the investigation of outliers also highlights a further noteworthy pattern. Cat. 19 pots and the FG 47 pottery stand form a mirrored group, each extremely concentrated at one site but absent from the other. This suggests the presence of distinct ceramic traditions, possibly reflecting the residence of different cultural groups or specialized activities in specific areas.

6. Conclusion

This dissertation examines the pottery assemblages from two contemporaneous Uruk-period istes in northern Syria, Jebel Aruda and Habuba Kabira-Süd. Both sites are thought to be uruk enclaves established during uruk exapsnion. Enabled by newly published excavation reports, this study challenges the traditional assumption of homogeneity among Uruk enclaves by comparing typology, frequency, and distribution of ceramic assemblages. The findings reveal both shared characteristics and significant differences, highlighting distinct cultural practices and interactions with local Chalcolithic traditions at each site.

The typological comparison identifies 58 shared ceramic forms, comprising 19 open shapes (such as BRBs, BTs, goblets, basins, ladles) and 39 closed shapes (including cooking pots, Ururk Red vessels, elongated jars with spouts, bottle shaped vessels, and installations like gutters). These forms align with typical Uruk-period assemblages, indicating broadly comparable functional repertoires associated with daily activities. However, frequency analysis reveals significant variation in their prevalence. At Jebel Aruda, the top five shared forms, excluding BRBs and BTs, are predominantly domestic, with Type 3 small goblets being the most numerous. In contrast, Habuba Kabira-Süd's top five included three prestige-related forms, notably Type 26 Uruk Red vessel, suggesting a stronger association with elite activities and potentially a higher position in the settlement hierarchy. This disparity challenges the characterization of Jebel Aruda as a religious or administrative center, where a greater presence of prestige objects might be anticipated. Beyond the top five, other shared forms also exhibit marked frequency differences. Distribution analysis further supports this pattern, demonstrating that while the two sites share many common ceramic forms, their patterns of use and concentration differ in significant ways. The identification of unique and highly localized forms not only highlights these difference but also points to questions about cultural practices and group identity of two sites within the Uruk expansion.

The analysis of unique ceramic forms further highlights the distinct identities of Jebel Aruda and Habuba Kabira-Süd while also exposing unexpected connections to local Late Chalcolithic traditions. Jebel Aruda produced eight unique forms, notebly Type 62 chaff-tempered high-neck pots and Type 66 hedgehog-shaped vessels. Habuba Kabira-Süd yielded 35 unique forms, with Type 67 pottery stands and Type 78 flat-bottom pots showing statistical significance and featuring a distinctive "pseudo-reserved slip" surface treatment applied to several unique forms such as Type 82 bottles. Many of Habuba Kabira-Süd's forms, particularly those in the "private production" Group IV, reflect local Late Chalcolithic influences. Importantly, however, Jebel Aruda also exhibits connections to local traditions. Like Habuba Kabira-Süd, Jebel Aruda includes chaff-tempered vessels and hybrid forms suggestive of cultural imitation or adoption. When hybrid forms are considered, the proportion of local elements at Jebel Aruda exceeds that at Habuba Kabira-Süd. Consequently, neither assemblage can be regarded as entirely "pure" Uruk. Furthermore, item-by-item distribution analysis of unique forms identifies mirrored pairs: Type 62 chaff-tempered pots at Jebel Aruda and Type 67 pottery stands at Habuba Kabira-Süd, each exclusive to one site and concentrated in specific areas. These patterns suggest specific, localized practices within the settlements.

In conclusion, the comparative analysis of pottery assemblages at Jebel Aruda and Habuba Kabira-Süd reveals a complex dynamic during the Uruk expansion in northern Syria. Beneath a veneer of typological similarity, significant differences in unique forms, frequency, and distribution patterns highlight distinct cultural identities and practices at each site. These insights call for a more refined understanding of Uruk-period enclaves, recognizing them as diverse communities rather than uniform settlements. This study, therefore, advances the understanding of these two Uruk sites and contributes to the broader narrative of the Uruk expansion. Nonetheless, many aspects of this phenomenon remain unclear, and a full explanation will depend on future archaeological discoveries and comprehensive analyses of materials beyond pottery.

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Appendix A

The distribution matrices are provided in the attached Excel file.

Appendix B

Table 1: DBI, Quantity, DII and Classification of the Jebel Aruda Pottery Assemblage by Abundance and Spatial Distribution

Object	DBI	Quantity	DII	Distribution Group
CAT 4	41	77	0.53	1
CAT 33	22	75	0.29	1
CAT 5	34	57	0.60	1
CAT 15	28	55	0.51	1
CAT 27	27	51	0.53	1
CAT 34	26	49	0.53	1
CAT 43	32	47	0.68	1
CAT 9	27	39	0.69	1
CAT 24	20	34	0.59	1
CAT 14	15	26	0.58	1
CAT 10	28	36	0.78	2
CAT 26	24	33	0.73	2
CAT 20	22	29	0.76	2
CAT 25	21	28	0.75	2
CAT 17	20	28	0.71	2
CAT 21	21	26	0.81	2
CAT 42	17	21	0.81	3
CAT 28	14	20	0.70	3
CAT 8	14	20	0.70	3
CAT 11	11	16	0.69	3
CAT 7	12	15	0.80	3

CAT 23	11	15	0.73	3
CAT 16	11	15	0.73	3
CAT 37	8	10	0.80	3
CAT 35	6	8	0.75	3
CAT 22	3	4	0.75	3
CAT 41	19	22	0.86	4
CAT 18	14	16	0.88	4
CAT 39	13	14	0.93	4
CAT 36	10	11	0.91	4
CAT 30	10	11	0.91	4
CAT 29	10	10	1.00	4
CAT 12	9	9	1.00	4
CAT 13	8	9	0.89	4
CAT 6	7	8	0.88	4
CAT 40	5	5	1.00	4
CAT 43 Hedgehog	3	3	1.00	4
CAT 38	2	2	1.00	4

Table 2: DBI, Quantity, DII and Classification of the Habuba Kabira-Süd Pottery Assemblage by Abundance and Spatial Distribution

Object	DBI	Quantity	DII	Distribution Group
FG 27	22	30	0.73	1
FG 31	21	40	0.53	1
FG 41	21	31	0.68	1
FG 12.1	8	10	0.80	1
FG 22.6	8	10	0.80	1
FG 29	7	9	0.78	1
FG 32	8	10	0.80	1
FG 36	12	19	0.63	1
FG 44	9	15	0.60	1
FG 1	12	14	0.86	2
FG 13	11	12	0.92	2
FG 2	6	7	0.86	2
FG 22.1	11	12	0.92	2
FG 22.2	6	7	0.86	2
FG 22.4	7	8	0.88	2
FG 22.5	9	11	0.82	2
FG 23	9	9	1.00	2

FG 24	10	11	0.91	2
FG 26	10	11	0.91	2
FG 3.2	10	10	1.00	2
FG 35.1	7	8	0.88	2
FG 35.2	7	8	0.88	2
FG 39.1	10	12	0.83	2
FG 40	15	15	1.00	2
FG 45	8	8	1.00	2
FG 47	7	7	1.00	2
FG 8	13	14	0.93	2
FG 10.1	3	4	0.75	3
FG 38	5	6	0.83	3
FG 48	3	4	0.75	3
FG 59	4	5	0.80	3
FG 63	5	6	0.83	3
FG 10.2	2	2	1.00	4
FG 11.2	2	2	1.00	4
FG 11.4	3	3	1.00	4
FG 12.2	3	3	1.00	4
FG 12.3	2	2	1.00	4
FG 12.4	4	4	1.00	4
FG 15	4	4	1.00	4
FG 20	4	4	1.00	4
FG 21	4	4	1.00	4
FG 22.3	5	5	1.00	4
FG 25	3	3	1.00	4
FG 28	2	2	1.00	4
FG 3.3	5	5	1.00	4
FG 37.2	3	3	1.00	4
FG 39.2	3	3	1.00	4
FG 42	3	3	1.00	4
FG 46	3	3	1.00	4
FG 49	2	2	1.00	4
FG 51	2	2	1.00	4
FG 52	2	2	1.00	4
FG 57	4	4	1.00	4
FG 58	4	4	1.00	4
FG 6	2	2	1.00	4
FG 7	2	2	1.00	4

Appendix C

Co-occurrence Analysis at Jebel Aruda

The Jaccard similarity coefficient was calculated to measure associations between object pairs. This index evaluates the similarity and diversity of datasets, calculated as the intersection size of two datasets divided by their union size. In our case, the index equals the number of rooms where pottery types A and B co-occur divided by the number of rooms where either pottery A or B occurs.

$$J(A,B)=\frac{|A\cap B|}{|A\cup B|}=\frac{Number\ of\ locations\ A\ and\ B\ co{-}occur}{Number\ of\ locations\ A\ or\ B\ appears}=\frac{N_{AB}}{N_{AA}+N_{BB}-N_{AB}}$$

Where:

N_{AB} is the number of rooms where both Cat. A and Cat. B appear (off-diagonal value)

N_{AA} is the number of rooms where Cat. A appears (diagonal value)

N_{BB} is the number of rooms where Cat. B appears (diagonal value)

The higher the index value, the more likely two categories co-occur. When J (A,B)=1, Cat. B always appears wherever Cat. A is present.

Table 1: Pottery Pairs with Jaccard Similarity Coefficients >0.25

Object 1	Object 2	Jaccard Index

CAT 25	CAT 24	0.375
CAT 26	CAT 25	0.351
CAT 27	CAT 34	0.349
CAT 27	CAT 26	0.326
CAT 4	CAT 9	0.310
CAT 34	CAT 21	0.308
CAT 3	CAT 5	0.307
CAT 9	CAT 25	0.300
CAT 4	CAT 34	0.293
CAT 4	CAT 15	0.290
CAT 3	CAT 4	0.289
CAT 15	CAT 27	0.286
CAT 27	CAT 7	0.286
CAT 20	CAT 24	0.286
CAT 21	CAT 23	0.286
CAT 25	CAT 23	0.286
CAT 4	CAT 27	0.283
CAT 5	CAT 27	0.283
CAT 43	CAT 26	0.280
CAT 34	CAT 26	0.279
CAT 4	CAT 26	0.276
CAT 27	CAT 24	0.275
CAT 5	CAT 26	0.275
CAT 9	CAT 26	0.273
CAT 5	CAT 21	0.271
CAT 43	CAT 9	0.269
CAT 4	CAT 10	0.267
CAT 5	CAT 20	0.265
CAT 33	CAT 24	0.265
CAT 15	CAT 7	0.263
CAT 26	CAT 24	0.263
CAT 8	CAT 16	0.261
CAT 9	CAT 7	0.257
CAT 26	CAT 18	0.257

A threshold of >0.3, representing the lowest index value in the top 10% of the data, was applied to identify strong pair relationships. A second threshold of >0.25 was used to detect “largest potential sets” where multiple pairs share weaker but meaningful relationships.

The data shows:

A. Strong pairs: {Cat. 25, Cat. 24}, {Cat. 26, Cat. 25}, {Cat. 27, Cat. 34}, {Cat. 27, Cat. 26}, {Cat. 4, Cat. 9}, {Cat. 34, Cat. 21}, {Cat. 3, Cat. 5}, {Cat. 9, Cat. 25}

B. Linkages among strong pairs helped identify two potential "object sets":{Cat. 25, Cat. 26, Cat. 27 and Cat. 34} and {Cat. 3, Cat. 4, Cat. 5, Cat. 9}

Within each set, more than half of the pairs show strong pairwise relationships, while the remaining pairs exhibit slightly weaker but still meaningful associations. The coefficients are listed below:

(1) Set 1: {Cat. 25, Cat. 26, Cat. 27 and Cat. 34}:

- J(Cat. 25, Cat. 26) = 0.351
- J(Cat. 25, Cat. 27) = 0.233 (weaker)
- J(Cat. 25, Cat. 34) = 0.244 (weaker)
- J(Cat. 26, Cat. 27) = 0.326
- J(Cat. 26, Cat. 34) = 0.279
- J(Cat. 27, Cat. 34) = 0.349

Cat. 21, though strongly linked with Cat. 34 (J=0.308), was not included in the group as its relationships with other categories were significantly weaker (J(Cat. 21, Cat. 25) = 0.15, J(Cat. 21, Cat. 26) = 0.136 and J(Cat. 21, Cat. 27) = 0.205).

Most of these sets concentrate in the southern area of the site. For example, in Room S11, two Cat. 25, three Cat. 26, one Cat. 27, and four Cat. 34 appear. In S12, one object of each four categories appears. Similar full sets also occur in Room S92, Pits S59 and S69, and Courtyard N58, with some incomplete sets occurring in many other southern rooms and two northern rooms (N8 and N18). In S11, a small storage room in the domestic quarter of the wealthiest Building SI on the site and where the curious hedgehog is also stored, the large quantity of the sets corresponds with the status of the building.

The appearance of one of each four categories in S12 (see Table 2), the reception room of Building SIII, strongly indicates the “set”. Does this mean that in the southern area, the liquid content stored in Cat. 25 and Cat. 26 and other dry food stored in Cat. 27 and Cat. 34 formed a “standard” reception service?

It is also interesting that N58, interpreted as an outdoor kitchen area by the excavator, is the only location in the northern area where a full Set 1 appears. The appearance of the set possibly indicates that N58 is either: (1) an important area where various storage and daily activities occurred, or (2) a normal kitchen area with rubbish dumps where the sets happened to gather. In either case, the uneven distribution of the sets in southern and northern areas indicates different “wealth statuses” across the site.

Table 2: Rooms with Full Set 1

	Category	Quantity	Pottery registration
S11	Cat. 25	2	JAF 51, JAF 53
	Cat. 26	3	JA 122, JAF 54, JAF 55
	Cat. 27	1	JA 926
	Cat. 34	4	JA 1640, JA 1648, JA 1650, JA 1655
S12	Cat. 25	1	JAF 226
	Cat. 26	1	JAF 239
	Cat. 27	1	JA 955
	Cat. 34	1	JA 1641
N58	Cat. 25	1	JAF 46
	Cat. 26	2	JA 1631, JAF 91
	Cat. 27	1	JA 238
	Cat. 34	1	JA 1656

(2) {Cat. 3, Cat. 4, Cat. 5, Cat. 9}:

- J(Cat. 3, Cat. 4) = 0.289
- J(Cat. 3, Cat. 5) = 0.307
- J(Cat. 3, Cat. 9) = 0.171 (weaker)
- J(Cat. 4, Cat. 5) = 0.250

- J(Cat. 4, Cat. 9) = 0.310
- J(Cat. 5, Cat. 9) = 0.241(weaker)

Cat. 25, though showing a relatively strong link with Cat. 9 (J=0.3), was excluded from the group due to weaker relationships with other categories (J(Cat. 25, Cat. 3)=0.186; J(Cat. 25, Cat. 4)=0.250; J(Cat. 25, Cat. 5)=0.245).

The excavators identified drinking sets comprising Cat. 3/5/9 based on pottery distribution patterns. Statistical analysis supports this observation while additionally revealing Cat. 4's strong pairing with Cat. 9 and moderately strong relationships with Cat. 3 and 5.

Table 3: The Location and Frequency of Pottery Sets

Set	Frequency	Room (Location)
3,4,5,9	7	S2/S3/S64/pit s59/Pit s69/N100/S114
3,4,5	4	S52/S92/S106/S11
4,5,9	4	S55/S84/S58/S70
3,4,9	3	S44/S57/S120
3,5,9	2	S27/S36

Again, most sets appear in the southern area, with only one northern room (N100, a storage room) containing all four categories (see Table 3). In the south, 4 of 8 buildings have complete sets: SI (S2/S3), SII (S64), SIV (Pits S59/69), and SVI (S114). Except for SVI, buildings SI/SII/SIV are all located on the lower terrace. SVI's unique position may indicate different functions or status compared to other upper-terrace buildings (SV/SVII).

The frequent appearance of round-bottomed Cat. 5 vessels in sets ({3,4,5,9}, {3,4,5}, {4,5,9}, {3,5,9}) suggests their essential role in eating/drinking activities. Unlike flat-based vessels that could be placed on surfaces, round-based ones required supports. The co-occurrence of both types implies they served different, non-interchangeable food purposes.